

8 October - 31 October

Office Mon-Fri 10:00-19:00. Saturday aur Sunday study days.

Roz ka ghadi, office day

06:00-07:20	1h 20m	Naya technical topic
07:20-07:35	15 min	Break. Agle sawal khol ke rakho
07:35-08:50	1h 15m	Technical practice, about 25 MCQ
08:50-10:00	1h 10m	Nikalna. 9:15 tak ghar se bahar
10:00-13:00	3h	Office kaam. Naya chapter nahi
13:00-13:20	20 min	Zaroori: aaj ki formula sheet + 6 MCQ
13:20-16:30	3h 10m	Office kaam
16:30-16:42	12 min	Zaroori: sheet band, formulas yaad
16:42-18:50	2h 8m	Office kaam
18:50-19:00	10 min	Zaroori: 5 formulas pocket card
19:00-20:00	1h	Ghar, khana, kapde
20:00-20:25	25 min	General Awareness
20:25-21:40	1h 15m	Doosra technical hissa + MCQ
21:40-21:50	10 min	Paani aur stretch
21:50-22:35	45 min	Reasoning
22:35-22:50	15 min	Error copy. Phir neend

Weekend

Shanivar: kareeb 7-8 ghante. Beech mein 12:00-14:30 aaram.

Ravivar: subah topic, 2 baje timed test, shaam ko analysis.

Strict rules

Paper-II merit hai: 100 sawal, 300 marks, har galat par -1. Andaza kam.

Paper-I shortlist hai: Reasoning 50, GA 50, CS 100, har galat par -0.25.

Tracker future slot lock karta hai. Time shuru hone se pehle tick nahi hota.

Jo slot nikal gaya aur tick nahi, woh Missed rehta hai jab tak complete na karo.

Ek hi notes aur ek error copy. 23:00 ke baad naya topic nahi.

Office ke teen slot zaroori hain: lunch formula, chai recall, nikalte hue 5 line.

Formula sheet ant ke pages par hai. Lunch par wahi din ki sheet.

24 din, ek nazar

Date	Topic	Padhai	MCQ
8 Oct Guruvar	Digital Logic	6h 27m	61
9 Oct Shukravar	Computer Organization	5h 57m	76
10 Oct Shanivar	C aur linear data structures	9h 20m	100
11 Oct Ravivar	Trees, heap, graph aur hafta test	7h	110
12 Oct Somvar	Algorithms: sort, search, hash	5h 57m	76
13 Oct Mangalvar	Greedy, DP, divide-and-conquer	5h 57m	76
14 Oct Budhvar	Graph algorithms	5h 57m	71
15 Oct Guruvar	OS: process, sync, deadlock	5h 57m	76
16 Oct Shukravar	OS numerical: schedule aur paging	5h 57m	71
17 Oct Shanivar	DBMS: ER, algebra, SQL	9h 20m	90
18 Oct Ravivar	Normalization, transactions, hafta test	6h 55m	115
19 Oct Somvar	Networks: layers aur data link	5h 57m	76
20 Oct Mangalvar	IP aur subnetting	5h 57m	76
21 Oct Budhvar	Transport aur application	5h 57m	76
22 Oct Guruvar	Theory of Computation	5h 57m	76
23 Oct Shukravar	Compiler Design	5h 57m	76
24 Oct Shanivar	Maths: discrete aur linear algebra	8h 45m	85
25 Oct Ravivar	Calculus, probability, hafta test	7h 20m	120
26 Oct Somvar	Revision: Logic, COA, DS	5h 57m	106
27 Oct Mangalvar	Revision: Algorithms aur OS	5h 57m	106
28 Oct Budhvar	Revision: DBMS aur Networks	5h 57m	106
29 Oct Guruvar	Revision: TOC, Compiler, Maths	5h 57m	96
30 Oct Shukravar	Paper-II mock	4h 52m	110
31 Oct Shanivar	Paper-I mock aur band	7h 50m	245

Padhai ka time office aur breaks ke bagair hai. MCQ us din ke technical + reasoning + GA ka target hai.

8 Oct Guruvar**Digital Logic**

Padhai 6h 27m | 61 technical 0 reasoning 0 GA

Aaj ka zaroori kaam raat 8 baje se 11:20 tak hai. Subah wale slots tabhi karo jab yeh din abhi subah se shuru ho. Office ja chuke ho to subah chhodo - raat wale char slots mein poora chapter hai. Dono mat ghisna.

Official syllabus lines

Paper-I + II 14.3.4 Digital Logic: Boolean algebra

Paper-I + II 14.3.4 Digital Logic: Combinational and sequential circuits

Paper-I + II 14.3.4 Digital Logic: Minimization

Paper-I + II 14.3.4 Digital Logic: Number representations and computer arithmetic (fixed and floating point)

Formula sheet for this dayDigital Logic: De Morgan = $(A+B)' = A'B'$ $(AB)' = A'+B'$. NAND is OR of complements. NOR is AND of complements.Digital Logic: XOR aur XNOR = $A \text{ xor } B = A'B + AB'$ $XNOR = AB + A'B'$. XOR is 1 when bits differ. XNOR is 1 when bits match.Digital Logic: Absorption = $A + AB = A$ $A(A+B) = A$ $A + A'B = A + B$. Drop the larger product term when one term contains the other.Digital Logic: 2's complement = Invert every bit, then add 1. Range of n bits: $-2^{(n-1)} \dots 2^{(n-1)}-1$. Unsigned range is $0 \dots 2^n - 1$.

One's complement is invert only.

Digital Logic: r's complement = $(r-1)$'s complement = $(r^n - 1) - N$. r's complement = $r^n - N$. For decimal, 9's then add 1 gives 10's complement.Digital Logic: Flip-flop next state = SR (SR=0): $Q+ = S + R'Q$. JK: $Q+ = JQ' + K'Q$. D: $Q+ = D$. T: $Q+ = T \text{ xor } Q$. SR = 11 is invalid. JK = 11 toggles. T = 1 toggles.Digital Logic: Counter aur MUX = n flip-flops, ripple up-counter: mod 2^n , $f_{\text{out}} = f_{\text{in}} / 2^n$. MUX: 2^n data, n select. Decoder: n to 2^n . A mod-10 counter needs extra reset logic. It is not a pure 4-bit free counter.

Digital Logic: K-map group = Group size must be 1, 2, 4, 8, ... Circle power-of-two 1s (or 0s for POS). Don't-care may join a group, never a group of its own. Adjacent cells differ by one bit, including wrap of the edges.

Digital Logic: IEEE-754 single = 1 sign, 8 exponent, 23 fraction. Bias 127. Normal value = $(-1)^s * (1.f) * 2^{(E-127)}$. E = 0 and f = 0 is zero. E = 255 and f = 0 is infinity. E = 255 and f != 0 is NaN. Double bias is 1023.**06:00-07:20****1h 20m****THEORY****Optional: Boolean aur K-map**

Sirf tab, jab subah free ho. Boolean laws, De Morgan, SOP/POS, 2/3/4 variable K-map, don't care. Number systems aur 1's, 2's complement.

07:20-07:35**15 min****BREAK****Break**

Optional subah ke saath hi. Raat wala plan is par depend nahi karta.

07:35-08:50**1h 15m****PRACTICE****Optional: 25 MCQ Digital Logic**

K-map aur complement. Subah kar liya ho to raat ka practice slot fir bhi rakho, par theory doosri baar mat padho.

25 technical MCQ**08:50-10:00****1h 10m****BREAK****Office ke liye nikalna**

9:15 tak nikalna. Aaj pehla din hai, bag mein sirf khaali error copy.

10:00-13:00**3h****OFFICE****Office, subah**

Kaam. 13:00 par Digital Logic ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20**20 min****OFFICE DESK****Lunch: Digital Logic**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Digital Logic.

6 technical MCQ

8 Oct Guruvar

Digital Logic (contd)

Padhai 6h 27m | 61 technical 0 reasoning 0 GA

13:20-16:30 3h 10m OFFICE

Office, dopahar

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK

Chai: formula band karke

12 minute, zaroori. Sheet band. Digital Logic mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 2h 8m OFFICE

Office, shaam

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 10 min OFFICE DESK

Nikalte hue 5 formulas

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Digital Logic.

19:00-20:00 1h BREAK

Ghar aur khana

Ek ghanta. 20:00 baje desk, phone dusre kamre.

20:00-21:15 1h 15m THEORY

Boolean, K-map, complements

Laws, De Morgan, SOP/POS, K-map 2 se 4 variable, don't care. Binary, octal, hex. Signed magnitude, 1's aur 2's complement. Yahi subah wale optional ka poora hissa hai.

21:15-21:25 10 min BREAK

Das minute

Paani. Agla hissa circuits ka hai.

21:25-22:25 1h THEORY

Circuits aur number representation

Half/full adder, MUX, decoder, encoder, comparator. Latch vs flip-flop. SR, JK, D, T ki characteristic equation. Ripple aur synchronous counter, shift register. Fixed vs floating point: sign, mantissa, exponent, bias.

22:25-23:05 40 min PRACTICE

30 MCQ, mix Digital Logic

K-map, flip-flop aur complement teeno se sawal hon. Last 15 minute sirf galat sawal. Reasoning aur GA aaj nahi - kal se roz shuru.

30 technical MCQ

23:05-23:20 15 min ERROR LOG

Paanch line aur neend

Error copy: De Morgan, 2's complement ke steps, JK/D/T equations, floating-point ke hisse. 23:20 ke baad band.

9 Oct Shukravar

Computer Organization

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Computer Organization and Architecture: Machine instructions and addressing modes

Paper-I + II 14.3.4 Computer Organization and Architecture: ALU, data-path and control unit

Paper-I + II 14.3.4 Computer Organization and Architecture: Instruction pipelining, pipeline hazards

Paper-I + II 14.3.4 Computer Organization and Architecture: Memory hierarchy: cache, main memory and secondary storage

Paper-I + II 14.3.4 Computer Organization and Architecture: I/O interface (interrupt and DMA mode)

Paper-I 14.2.1 General Intelligence & Reasoning: Analogies

Paper-I 14.2.2 General Awareness: History, India and its neighboring countries

Formula sheet for this day

Computer Organization: CPU time = CPU time = IC * CPI * clock cycle. Clock cycle = $1 / \text{frequency}$. IC is instruction count. A lower CPI or a faster clock both cut time.

Computer Organization: Pipeline speedup = No stalls, n instructions, k stages: $S = (n*k) / (k + n - 1)$. Large n: S approaches k. One stall cycle drops the ideal throughput. Efficiency = S / k .

Computer Organization: Stall CPI = Ideal pipeline CPI = 1. Effective CPI = $1 + \text{stall cycles per instruction}$. Data hazard: forwarding removes many, not all, stalls. Control hazard: branch delay or flush.

Computer Organization: Cache AMAT = AMAT = Hit time + Miss rate * Miss penalty. Miss rate = $1 - \text{hit rate}$. Two-level: $\text{AMAT} = T_{L1} + M1*(T_{L2} + M2*T_{\text{mem}})$.

Computer Organization: Hit-rate EAT = EAT = $h*T_{\text{cache}} + (1-h)*T_{\text{memory}}$. If every access also pays the cache lookup: $\text{EAT} = h*T_c + (1-h)*(T_c + T_m)$. Use the form the question states.

Computer Organization: Cache address split = Offset = $\log_2(\text{block bytes})$. Direct index = $\log_2(\text{blocks})$. Set index = $\log_2(\text{sets})$. Tag = address bits - index - offset. Sets = blocks / ways. Cache bytes = sets * ways * block bytes.

Computer Organization: Effective address = Direct: EA = address field. Indirect: EA = M[address]. Indexed: EA = index + address.

Relative: EA = PC + offset. Register-indirect: EA = register. Immediate has no memory EA. The operand sits in the instruction.

Computer Organization: Interrupt vs DMA = Interrupt: CPU saves context and runs the handler. DMA: device moves a block; CPU is told at the end. Cycle stealing takes the bus for single transfers. Burst mode holds the bus for the block.

06:00-07:20 1h 20m THEORY

Instructions aur addressing modes

Instruction cycle. Opcode aur operand. Modes: immediate, direct, indirect, register, register-indirect, indexed, relative. ALU ka kaam. Hardwired vs microprogrammed control. PC, IR, MAR, MBR ka role, detail architecture nahi.

07:20-07:35 15 min BREAK

Chhoti break

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE

25 MCQ - mode identify karo

Ek instruction dekar mode poochha jaata hai. Galat ho to definition ek line mein error copy. Aakhri 20 minute sirf review.

25 technical MCQ

08:50-10:00 1h 10m BREAK

Office ke liye nikalna

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE

Office, subah

Kaam. 13:00 par Computer Organization ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK

Lunch: Computer Organization

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Computer Organization.

6 technical MCQ

9 Oct Shukravar**Computer Organization (contd)**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

13:20-16:30**3h 10m****OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42**12 min****OFFICE DESK****Chai: formula band karke**

12 minute, zaroori. Sheet band. Computer Organization mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50**2h 8m****OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00**10 min****OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Computer Organization.

19:00-20:00**1h****BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25**25 min****GA****Itihaas, sirf timeline**

IVC, Vedic, Maurya, Gupta, Delhi Sultanate, Mughals. Har ek ka ek shasak aur ek kaam. Kahani nahi.

10 GA**20:25-21:40****1h 15m****THEORY****Pipeline, cache, interrupt, DMA**

Pehle 45 minute: ideal speedup, stall, data/control/structural hazard, forwarding. Cache: direct, set-associative, fully associative, hit rate, write-through vs write-back, effective access time. Interrupt vs DMA. Aakhri 30 minute: 15 MCQ.

15 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Coding-decoding**

Letter shift aur number coding, 20 sawal. Tees second mein pattern na dikhe to aage badho, ant mein lautna.

20 reasoning**22:35-22:50****15 min****ERROR LOG****Teen formula**

Likh kar so jao: ideal speedup, hit rate, effective memory access time. 23:00 tak light band.

10 Oct Shanivar**C aur linear data structures**

Padhai 9h 20m | 65 technical 20 reasoning 15 GA

Official syllabus lines

Paper-I + II 14.3.4 Programming and Data Structures: Programming in C

Paper-I + II 14.3.4 Programming and Data Structures: Recursion

Paper-I + II 14.3.4 Programming and Data Structures: Arrays, stacks, queues, linked lists

Paper-I 14.2.1 General Intelligence & Reasoning: Similarities

Paper-I 14.2.2 General Awareness: Culture

Formula sheet for this day

C, arrays, stacks, queues, lists: Array address = 1D: $\text{base} + i * \text{size}$. Row-major $A[i][j] = \text{base} + (i * \text{cols} + j) * \text{size}$. Column-major: $\text{base} + (j * \text{rows} + i) * \text{size}$. C uses row-major and 0-based indexes.

C, arrays, stacks, queues, lists: Pointer step = $p + k$ moves k elements, not k bytes. Byte move = $k * \text{sizeof}(*p)$. $a[i]$ is $*(a+i)$. $\&a[i]$ is $a+i$.

C, arrays, stacks, queues, lists: Stack aur queue = Stack: push and pop at the same end, $O(1)$. Circular queue: front and rear mod capacity, $O(1)$. Infix to postfix uses a stack. BFS uses a queue. DFS uses a stack.

C, arrays, stacks, queues, lists: Linked list = Insert or delete at a known node: $O(1)$. Search: $O(n)$. Extra memory: one pointer per node. Doubly linked delete of a known node does not need the previous walk.

C, arrays, stacks, queues, lists: Recursion cost = Time follows the recurrence of the calls. Depth of the call stack is the deepest chain, often $O(n)$ or $O(\log n)$. A missing base case overflows the stack. Tail recursion can reuse the frame.

07:00-09:00 2h THEORY**C jo MCQ mein aata hai**

Pointers, pointer arithmetic, arrays, strings, storage class, call by value. Recursion tree aur base case. Output-based sawal ka andaaz samjho, poori language nahi.

09:00-09:30 30 min BREAK**Nashta**

Aadha ghanta. Phone yahin khatam.

09:30-11:30 2h PRACTICE**30 MCQ - C output aur recursion**

Galat output ko dry-run ke saath error copy mein likho. Sirf jawab mark karke aage mat badho.

30 technical MCQ**11:30-12:10 40 min REASONING****Direction aur ranking**

20 sawal. Distance-direction mein diagram bina solve kiye jawab mat chuno.

20 reasoning**12:10-14:30 2h 20m BREAK****Khana aur aaram**

Yeh aaram plan ka hissa hai. 14:30 se pehle agla chapter mat kholo.

14:30-16:30 2h THEORY**Array, stack, queue, linked list**

Insert aur delete ki complexity. Stack se infix/postfix ka idea. Circular queue. Singly, doubly, circular list par insertion ke cases. Code yaad karne ki jagah operation yaad karo.

16:30-17:00 30 min GA**Bharat ka physical map**

Nadiyan, darre, mittiyan, monsoon. Das se pandrah facts, ek baar likh lo.

15 GA**17:00-18:40 1h 40m PRACTICE****35 MCQ - stack, queue, list**

Complexity wale sawal alag se mark karo. Jo structure confuse ho, uska ek chhota diagram banao.

35 technical MCQ

10 Oct Shanivar

C aur linear data structures (contd)

Padhai 9h 20m | 65 technical 20 reasoning 15 GA

18:40-19:10 30 min ERROR LOG

Complexity ki ek table

Array, stack, queue, singly aur doubly list ke insert/delete/search. Iske baad aaj band. Raat ko naya topic nahi.

11 Oct Ravivar**Trees, heap, graph aur hafta test**

Padhai 7h | 70 technical 25 reasoning 15 GA

Official syllabus lines

Paper-I + II 14.3.4 Programming and Data Structures: Trees, binary search trees

Paper-I + II 14.3.4 Programming and Data Structures: Binary heaps

Paper-I + II 14.3.4 Programming and Data Structures: Graphs

Paper-I 14.2.1 General Intelligence & Reasoning: Differences

Paper-I 14.2.2 General Awareness: Geography

Formula sheet for this day

Trees, heaps, graphs: Binary tree counts = Max nodes at level L (root level 0) = 2^L . Perfect tree height h : nodes = $2^{(h+1)} - 1$. Full tree: leaves = internal nodes + 1. Height of a single node is 0 in this sheet. Some books use 1. Match the question.

Trees, heaps, graphs: BST = Left subtree < node < right subtree. Search, insert, delete: $O(h)$. Balanced $h = \text{floor}(\log_2 n)$. Skewed $h = n-1$. Inorder of a BST is sorted order.

Trees, heaps, graphs: Binary heap index = 0-based: parent $\text{floor}((i-1)/2)$, left $2i+1$, right $2i+2$. Height $\text{floor}(\log_2 n)$. Build $O(n)$. Extract $O(\log n)$. Min-heap: parent $\leq$ children. Heap is a complete binary tree.

Trees, heaps, graphs: Graph storage = Adjacency matrix: $O(V^2)$ space, edge check $O(1)$. Adjacency list: $O(V+E)$ space, edge check $O(\text{degree})$. Undirected edge appears twice in the list, once in the matrix triangle.

08:00-10:00 2h THEORY**Tree, BST, heap, graph**

Inorder, preorder, postorder. BST insert, search, delete. AVL sirf balance factor tak. Binary heap insert/delete. Graph: adjacency list vs matrix, degree. Traversal ka code nahi, order yaad hai.

10:00-10:20 20 min BREAK**Break**

Chai. Test ka set alag rakhna, abhi use mat kholo.

10:20-11:40 1h 20m PRACTICE**30 MCQ - tree, BST, heap, graph**

Traversal wale sawal haath se banao. Sirf option padh kar mat chuno.

30 technical MCQ**11:40-14:00 2h 20m BREAK****Khana aur aaram**

Test dopehar ko hai. Is aaram ko skip karke subah ka topic dobara mat padho.

14:00-14:40 40 min MOCK**Mini test: 40 sawal, 40 minute**

Digital Logic + COA + DS. Timer. Phone dusre kamre. Jo nahi aata chhodo, negative -0.25 maan kar.

40 technical MCQ**14:40-15:40 1h ERROR LOG****Test ka analysis**

Har galat par topic tag. Teen se zyada galat wale topic ka naam kal ke error copy ke upar likho.

15:40-16:25 45 min REASONING**Non-verbal**

Figure series, mirror, water image, paper folding. 25 sawal.

25 reasoning**16:25-17:10 45 min GA****Hafte ka static mix**

Polity, itihaas, geo se 15 sawal. Naya chapter nahi.

15 GA

11 Oct Ravivar

Trees, heap, graph aur hafta test (contd)

Padhai 7h | 70 technical 25 reasoning 15 GA

17:10-17:40 30 min ERROR LOG

Hafta band

Error copy band karo. Kal Algorithms hai - sorting ki table wale notes sirf nikaal ke rakh do, padhna kal subah.

12 Oct Somvar**Algorithms: sort, search, hash**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Algorithms: Searching, sorting, hashing

Paper-I + II 14.3.4 Algorithms: Asymptotic worst-case time and space complexity

Paper-I 14.2.1 General Intelligence & Reasoning: Space visualization

Paper-I 14.2.2 General Awareness: Economic scene

Formula sheet for this day

Searching, sorting, hashing: Binary search = Comparisons about $\text{floor}(\log_2 n) + 1$. Needs a sorted array. Time $\Theta(\log n)$. Lower bound of comparison sort is $\Omega(n \log n)$.

Searching, sorting, hashing: Sorting cost = Insertion, selection, bubble: worst $\Theta(n^2)$. Merge: $\Theta(n \log n)$, extra $O(n)$. Heap: $\Theta(n \log n)$, in place. Quick: average $\Theta(n \log n)$, worst $\Theta(n^2)$. Insertion best case $\Theta(n)$ on sorted input. Merge and heap are stable? Merge is stable. Heap and quick are not.

Searching, sorting, hashing: Stability = Stable: equal keys keep their input order. Merge and insertion are stable. Selection, heap, and quick are not. Counting sort is stable if the prefix scan fills from the end.

Searching, sorting, hashing: Counting aur radix = Counting: $O(n + k)$ time, k is the key range. Radix: $O(d(n + k))$ for d digits. Use when keys are integers in a known range, not for arbitrary comparisons.

Searching, sorting, hashing: Hash load = $\alpha = n / m$. Chaining, unsuccessful search about $1 + \alpha$. Successful about $1 + \alpha/2$. Open addressing degrades as α approaches 1. Keep α well below 1.

06:00-07:20 1h 20m THEORY**Search, sort, asymptotic notation**

Linear aur binary search. Bubble, selection, insertion, merge, quick, heap, counting, radix: best, average, worst, stable, in-place. Big-O, Omega, Theta. Simple recurrence sirf merge sort jitni.

07:20-07:35 15 min BREAK**Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE**25 MCQ - sorting identify karo**

Best/worst/stable wale sawal. Galat sort ka ek cell error copy mein. Comparison sort ki lower bound yaad rakhna.

25 technical MCQ**08:50-10:00 1h 10m BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE**Office, subah**

Kaam. 13:00 par Searching, sorting, hashing ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK**Lunch: Searching, sorting, hashing**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Searching, sorting, hashing.

6 technical MCQ**13:20-16:30 3h 10m OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK**Chai: formula band karke**

12 minute, zaroori. Sheet band. Searching, sorting, hashing mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

12 Oct Somvar**Algorithms: sort, search, hash (contd)**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

16:42-18:50**2h 8m****OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00**10 min****OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Searching, sorting, hashing.

19:00-20:00**1h****BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25**25 min****GA****Economy, definitions**

GDP, inflation, repo, reverse repo, CRR, budget, RBI ke kaam. Das definitions, ek line each.

10 GA**20:25-21:40****1h 15m****THEORY****Hashing, phir 15 MCQ**

Hash function, chaining, open addressing, collision, load factor. Primary clustering ka naam. 45 minute theory, 30 minute mein 15 MCQ search+sort+hash.

15 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Syllogism**

Do statements, Venn se. 20 sawal. 'Some not' wale cases alag se dhyaan.

20 reasoning**22:35-22:50****15 min****ERROR LOG****Sort table yaad se**

Bina notes ke 8 sorts ki teen complexities. Jo atak jaaye sirf wahi kal subah se pehle dekho. 23:00 band.

13 Oct Mangalvar**Greedy, DP, divide-and-conquer**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Algorithms: Algorithm design techniques: greedy, dynamic programming and divide-and-conquer

Paper-I 14.2.1 General Intelligence & Reasoning: Problem-solving

Paper-I 14.2.2 General Awareness: General polity

Formula sheet for this day

Greedy, DP, divide-and-conquer: Master theorem = $T(n) = a T(n/b) + f(n)$. Let $c = \log_b(a)$. If $f = O(n^{c-e})$: $\Theta(n^c)$. If $f = \Theta(n^c)$: $\Theta(n^c \log n)$. If $f = \Omega(n^{c+e})$ and $a f(n/b) \leq k f(n)$ for $k < 1$: $\Theta(f(n))$. e is a positive constant. Merge sort: $a=2$, $b=2$, $f=n$, $c=1$, case 2.

Greedy, DP, divide-and-conquer: Divide and conquer = Split, solve, combine. Binary search, merge sort, quicksort, Strassen. Cost is the recurrence, then Master theorem. Not every split is equal. Quicksort's worst split is 1 and $n-1$.

Greedy, DP, divide-and-conquer: Greedy choice = Feasible choice that looks best now, plus optimal substructure. Fractional knapsack: sort by value/weight. 0/1 knapsack is not greedy. It is DP. Activity selection: pick the finish-earliest.

Greedy, DP, divide-and-conquer: DP shape = Optimal substructure and overlapping subproblems. 0/1 knapsack: $dp[i][w] = \max(dp[i-1][w], v_i + dp[i-1][w-w_i])$. Fibonacci naive recursion is exponential. With a table it is $O(n)$.

06:00-07:20**1h 20m****THEORY****Greedy**

Fractional knapsack, activity selection, Huffman ka idea. Greedy kab fail hota hai, yeh 0/1 knapsack se farq. Proof nahi, choice ki property.

07:20-07:35**15 min****BREAK****Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50**1h 15m****PRACTICE****20 MCQ - greedy choice**

Kaun sa problem greedy hai, kaun DP. Galat classification error copy mein.

20 technical MCQ**08:50-10:00****1h 10m****BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00**3h****OFFICE****Office, subah**

Kaam. 13:00 par Greedy, DP, divide-and-conquer ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20**20 min****OFFICE DESK****Lunch: Greedy, DP, divide-and-conquer**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Greedy, DP, divide-and-conquer.

6 technical MCQ**13:20-16:30****3h 10m****OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42**12 min****OFFICE DESK****Chai: formula band karke**

12 minute, zaroori. Sheet band. Greedy, DP, divide-and-conquer mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50**2h 8m****OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

13 Oct Mangalvar

Greedy, DP, divide-and-conquer (contd)

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

18:50-19:00 10 min OFFICE DESK

Nikalte hue 5 formulas

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Greedy, DP, divide-and-conquer.

19:00-20:00 1h BREAK

Ghar aur khana

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25 25 min GA

Rozmarra science

Units, vitamins, common diseases, light aur sound ke basic facts. Das sawal.

10 GA

20:25-21:40 1h 15m THEORY

DP, Master theorem, 20 MCQ

0/1 knapsack, LCS, LIS - table padhna aur bharna. Matrix chain sirf idea. Master theorem: a, b, k ke teen cases, lamba proof nahi. Aakhri 25 minute 20 MCQ.

20 technical MCQ

21:40-21:50 10 min BREAK

Das minute

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35 45 min REASONING

Blood relation

20 sawal. Generation ka diagram. 'Only son of' wale sentences par ruko.

20 reasoning

22:35-22:50 15 min ERROR LOG

Ek DP table khud

LCS ya knapsack ka ek chhota table bina notes ke. 23:00 band.

14 Oct Budhvar

Graph algorithms

Padhai 5h 57m | 46 technical 15 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Algorithms: Graph traversals, minimum spanning trees, shortest paths

Paper-I 14.2.1 General Intelligence & Reasoning: Analysis

Paper-I 14.2.2 General Awareness: Scientific research

Formula sheet for this day

Graph algorithms: BFS aur DFS = Time $O(V+E)$ on a list. BFS gives fewest edges from the source in an unweighted graph. DFS gives finishing times. Topological sort: decreasing finish time, and only on a DAG.

Graph algorithms: MST = Kruskal: sort edges, add if no cycle, $O(E \log E)$. Prim: grow one tree, $O(E \log V)$ with a heap. MST has V-1 edges. Cut property: lightest edge across a cut is in some MST. Negative weights are allowed. Negative cycles are not a concept here.

Graph algorithms: Dijkstra = Non-negative weights. Heap version $O((V+E) \log V)$. $\text{Dist}[v] = \min(\text{dist}[v], \text{dist}[u] + w(u,v))$ when u is settled. A negative edge can make a settled node wrong. Use Bellman-Ford then.

Graph algorithms: Bellman-Ford aur Floyd = Bellman-Ford: relax all edges V-1 times, $O(VE)$. One more relax means a negative cycle. Floyd: $\text{dp}[k][i][j] = \min(\text{dp}[i][j], \text{dp}[i][k] + \text{dp}[k][j])$, $O(V^3)$. Floyd allows negative edges, not a negative cycle.

06:00-07:20 1h 20m THEORY

BFS aur DFS

Queue vs stack. Connected components, cycle detection undirected aur directed. Complexity adjacency list par.

07:20-07:35 15 min BREAK

Chhoti break

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE

20 MCQ - traversal

Chhote graph par BFS/DFS order haath se. Starting vertex dhyan se.

20 technical MCQ**08:50-10:00 1h 10m BREAK**

Office ke liye nikalna

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE

Office, subah

Kaam. 13:00 par Graph algorithms ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK

Lunch: Graph algorithms

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Graph algorithms.

6 technical MCQ**13:20-16:30 3h 10m OFFICE**

Office, dopahar

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK

Chai: formula band karke

12 minute, zaroori. Sheet band. Graph algorithms mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 2h 8m OFFICE

Office, shaam

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

14 Oct Budhvar

Graph algorithms (contd)

Padhai 5h 57m | 46 technical 15 reasoning 10 GA

18:50-19:00 10 min OFFICE DESK

Nikalte hue 5 formulas

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Graph algorithms.

19:00-20:00 1h BREAK

Ghar aur khana

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25 25 min GA

Current affairs ka tarika

Pichhle teen mahine ke 10 headings chuno. Aaj 4 headings, har ek par do line. Roz isi list mein se do revise karoge.

10 GA

20:25-21:40 1h 15m PRACTICE

MST, shortest path, topo sort

Prim aur Kruskal, cycle check. Dijkstra negative edge par nahi. Bellman-Ford ka idea. Topological sort. Phir 20 MCQ mix: sort + DP + graph.

20 technical MCQ

21:40-21:50 10 min BREAK

Das minute

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35 45 min REASONING

Linear seating

15 sawal ya do chhote sets. Poora arrangement likho, option se ulta mat jao.

15 reasoning

22:35-22:50 15 min ERROR LOG

Graph complexity

BFS, DFS, Prim, Kruskal, Dijkstra ki time complexity ek line each. 23:00 band.

15 Oct Guruvar**OS: process, sync, deadlock**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Operating System: System calls, processes, threads

Paper-I + II 14.3.4 Operating System: Inter-process communication

Paper-I + II 14.3.4 Operating System: Concurrency and synchronization

Paper-I + II 14.3.4 Operating System: Deadlock

Paper-I 14.2.1 General Intelligence & Reasoning: Judgment

Paper-I 14.2.2 General Awareness: India and its neighboring countries

Formula sheet for this day

Processes, sync, deadlock: TAT , waiting, response = TAT = completion - arrival. Waiting = TAT - burst. Response = first CPU start - arrival. CPU utilization = busy time / total time. Throughput = jobs finished / time.

Processes, sync, deadlock: Scheduling pick = FCFS: simple, convoy. SJF: least average waiting if all arrive together, non-preemptive. SRTF: preemptive SJF. RR: quantum q , more switches as q shrinks. Priority can starve unless aging raises the waiting job.

Processes, sync, deadlock: Banker's need = $Need[i] = Max[i] - Allocation[i]$. Safe if a sequence exists: Work starts as Available, and each picked process has $Need \leq Work$, then $Work += Allocation$. An unsafe state is not a deadlock yet. It might become one.

Processes, sync, deadlock: Deadlock four = Mutual exclusion, hold and wait, no preemption, circular wait. All four together. Break any one condition and deadlock cannot form. Prevention vs avoidance vs detection.

Processes, sync, deadlock: Semaphore = Wait (P): while $S \leq 0$ loop; $S = S - 1$. Signal (V): $S = S + 1$. Binary semaphore is 0 or 1.

Counting semaphore tracks a pool. Mutex lock is not a signal to a different process by itself. A semaphore can.

06:00-07:20 1h 20m THEORY**Process, thread, IPC**

System call kya hai. Process vs thread. PCB. States: new, ready, running, waiting, terminated. Shared memory vs message passing.

07:20-07:35 15 min BREAK**Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE**20 MCQ - process aur thread**

State transition wale sawal diagram se. Galat transition likho.

20 technical MCQ**08:50-10:00 1h 10m BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE**Office, subah**

Kaam. 13:00 par Processes, sync, deadlock ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK**Lunch: Processes, sync, deadlock**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Processes, sync, deadlock.

6 technical MCQ**13:20-16:30 3h 10m OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

15 Oct Guruvar**OS: process, sync, deadlock (contd)**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

16:30-16:42 **12 min** **OFFICE DESK****Chai: formula band karke**

12 minute, zaroori. Sheet band. Processes, sync, deadlock mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 **2h 8m** **OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 **10 min** **OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Processes, sync, deadlock.

19:00-20:00 **1h** **BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25 **25 min** **GA****Parliament aur emergency**

Lok Sabha, Rajya Sabha, President, PM. Money bill vs ordinary. National, state, financial emergency ka article number agar yaad ho, warna naam aur farq.

10 GA**20:25-21:40** **1h 15m** **THEORY****Semaphore aur deadlock**

Race, critical section, mutex, binary aur counting semaphore. Producer-consumer ka idea. Deadlock ki 4 conditions. Prevention, avoidance, Banker's ka concept, detection. 20 MCQ.

20 technical MCQ**21:40-21:50** **10 min** **BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35 **45 min** **REASONING****Inequality**

20 sawal. Symbols ko chain banao, yaad se mat chuno.

20 reasoning**22:35-22:50** **15 min** **ERROR LOG****Deadlock ki chaar lines**

Mutual exclusion, hold and wait, no preemption, circular wait. Apne shabdon mein. 23:00 band.

16 Oct Shukravar**OS numerical: schedule aur paging**

Padhai 5h 57m | 46 technical 15 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Operating System: CPU and I/O scheduling

Paper-I + II 14.3.4 Operating System: Memory management and virtual memory

Paper-I + II 14.3.4 Operating System: File systems

Paper-I 14.2.1 General Intelligence & Reasoning: Decision-making

Paper-I 14.2.2 General Awareness: Current events

Formula sheet for this day

Memory and file systems: Page bits = Offset bits = $\log_2(\text{page size})$. Page number = logical address bits - offset bits. Frames = physical bytes / page size. Logical pages and physical frames are the same size.

Memory and file systems: TLB EAT = Single-level page table: $EAT = h \cdot (T_{tlb} + T_{mem}) + (1-h) \cdot (T_{tlb} + 2 \cdot T_{mem})$. A miss reads the page table in memory, then the data. Two memory reads.

Memory and file systems: Page fault time = $EAT = (1-p) \cdot \text{memory access} + p \cdot \text{page-fault service}$. p is the fault probability. Service includes disk. It dominates memory.

Memory and file systems: Replacement = FIFO can show Belady: more frames, more faults. Optimal: replace the page used farthest in the future. LRU: replace the least recently used. Optimal needs the future. It is a yardstick, not an online policy.

Memory and file systems: Fragmentation = Paging: internal fragment in the last page only. Segmentation: external fragment between variable segments. Compaction moves segments. Paging does not need it for external holes.

Memory and file systems: File allocation = Contiguous: direct, external fragment. Linked: no external fragment, slow random.

Indexed: index block, random access, one extra read. Unix inode: direct blocks, then single, double, triple indirect.

06:00-07:20**1h 20m****THEORY****Memory ka nakshe**

Paging, segmentation, internal vs external fragmentation. Page table, frame. Virtual memory aur page fault ka matlab. File allocation contiguous, linked, indexed - sirf farq.

07:20-07:35**15 min****BREAK****Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50**1h 15m****PRACTICE****15 MCQ - memory concepts**

Fragmentation aur page fault wale seedhe sawal. Numerical raat ko hai.

15 technical MCQ**08:50-10:00****1h 10m****BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00**3h****OFFICE****Office, subah**

Kaam. 13:00 par Memory and file systems ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20**20 min****OFFICE DESK****Lunch: Memory and file systems**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Memory and file systems.

6 technical MCQ**13:20-16:30****3h 10m****OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16 Oct Shukravar

OS numerical: schedule aur paging (contd)

Padhai 5h 57m | 46 technical 15 reasoning 10 GA

16:30-16:42 12 min OFFICE DESK

Chai: formula band karke

12 minute, zaroori. Sheet band. Memory and file systems mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 2h 8m OFFICE

Office, shaam

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 10 min OFFICE DESK

Nikalte hue 5 formulas

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Memory and file systems.

19:00-20:00 1h BREAK

Ghar aur khana

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25 25 min GA

Aadhunik Bharat, saal

1857 se 1947. Andolan aur saal. Das events, kahani nahi.

10 GA

20:25-21:40 1h 15m PRACTICE

Gantt chart aur page replacement

FCFS, SJF, SRTF, Priority, Round Robin - kam se kam 4 schedule. Waiting time aur turnaround. FIFO, LRU, Optimal - 3 numerical. Phir jitna time bache 10 MCQ. Calculator ki aadat mat daalo.

25 technical MCQ

21:40-21:50 10 min BREAK

Das minute

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35 45 min REASONING

Do chhote puzzle

Floor ya box, 15 sawal. Ek puzzle 20 minute se aage mat kheecho.

15 reasoning

22:35-22:50 15 min ERROR LOG

Ek galat Gantt dubara

Jo schedule galat hua, bina notes ke ek baar aur. 23:00 band.

17 Oct Shanivar**DBMS: ER, algebra, SQL**

Padhai 9h 20m | 55 technical 20 reasoning 15 GA

Official syllabus lines

Paper-I + II 14.3.4 Databases: ER-model

Paper-I + II 14.3.4 Databases: Relational model: relational algebra, tuple calculus, SQL

Paper-I 14.2.1 General Intelligence & Reasoning: Visual memory

Paper-I 14.2.2 General Awareness: Environment around you and its application to society

Formula sheet for this day

ER, algebra, SQL, normal forms: Relational algebra = sigma select, pi project, rho rename, union, minus, cartesian product, join.

Select keeps rows. Project keeps columns and removes duplicate rows.

ER, algebra, SQL, normal forms: Keys = Superkey uniquely identifies. Candidate key is a minimal superkey. Primary key is one chosen candidate. Foreign key references a candidate key. Prime attribute: member of some candidate key.

ER, algebra, SQL, normal forms: Normal forms = 1NF: atomic values. 2NF: no partial dependency of a non-prime on a candidate key. 3NF: for $X \rightarrow A$, X is a superkey or A is prime. BCNF: for every nontrivial $X \rightarrow A$, X is a superkey. BCNF implies 3NF. 3NF does not imply BCNF. A table can be 3NF and still have overlapping keys.ER, algebra, SQL, normal forms: Lossless join = Decomposition of two parts is lossless if $R_1 \cap R_2$ is a key of R_1 or of R_2 .

Dependency preserving: the union of the projected FDs is equivalent to F. Lossless and preserving are different tests.

ER, algebra, SQL, normal forms: SQL groups = WHERE filters rows before groups. HAVING filters groups. SELECT aggregates need GROUP BY of the other selected columns. COUNT(*) counts rows. COUNT(col) skips NULL. DISTINCT inside COUNT counts uniques.

07:00-09:00 2h THEORY**ER, keys, relational algebra**

Entity, attribute, relationship, cardinality. Super, candidate, primary, foreign key. Select, project, union, difference, cartesian, theta/equi/natural/outer join, division.

09:00-09:30 30 min BREAK**Nashta**

Aadha ghanta.

09:30-11:30 2h PRACTICE**25 MCQ - algebra**

Query ka result chhota table bana kar nikalo. Division wale sawal skip mat karo, ek example samajh lo.

25 technical MCQ**11:30-12:10 40 min REASONING****Clock aur calendar**

20 sawal. Angle of clock ka formula ek baar likh lo, phir sawal.

20 reasoning**12:10-14:30 2h 20m BREAK****Khana aur aaram**

14:30 se pehle SQL mat kholo.

14:30-16:30 2h THEORY**SQL jo output poochhta hai**

SELECT, WHERE, GROUP BY, HAVING, JOIN, nested query, view, NULL. Clause ka order. HAVING vs WHERE.

16:30-17:00 30 min GA**Kala aur sanskriti**

Nritya, tyohar, UNESCO sites. Ek chhoti list, 15 facts.

15 GA**17:00-18:40 1h 40m PRACTICE****30 MCQ - SQL output**

Query likho ya table par chhalao, option se andaza mat lagao. NULL wale sawal alag tag.

30 technical MCQ

17 Oct Shanivar

DBMS: ER, algebra, SQL (contd)

Padhai 9h 20m | 55 technical 20 reasoning 15 GA

18:40-19:10 30 min ERROR LOG

Teen galat query dubara

Bina options ke result likho. Aaj yahin band.

18 Oct Ravivar**Normalization, transactions, hafta test**

Padhai 6h 55m | 75 technical 20 reasoning 20 GA

Official syllabus lines

Paper-I + II 14.3.4 Databases: Integrity constraints, normal forms

Paper-I + II 14.3.4 Databases: File organization, indexing (B and B+ trees)

Paper-I + II 14.3.4 Databases: Transactions and concurrency control

Paper-I 14.2.1 General Intelligence & Reasoning: Discrimination

Paper-I 14.2.2 General Awareness: Everyday observation in its scientific aspect

Formula sheet for this day

Indexing, transactions, concurrency: B and B+ tree = Order-m node holds up to m children. Search $O(\log n)$. B+ tree stores records only in the leaves, and leaves are linked. A B+ leaf scan is the range query. Internal nodes are a sparse directory.

Indexing, transactions, concurrency: Index kind = Dense: one entry per record. Sparse: one entry per block, file must be ordered.

Primary index is on the ordering key. Secondary index is on another field. Clustering index can be on a non-unique ordered field.

Indexing, transactions, concurrency: ACID = Atomic: all or nothing. Consistent: constraints hold. Isolated: concurrent effect matches some serial order. Durable: commit survives a crash. A schedule can be atomic per transaction and still not serializable.

Indexing, transactions, concurrency: Serializability = Conflict: two ops, different transactions, same item, at least one write. Conflict serializable iff the precedence graph has no cycle. Conflict serializable implies view serializable. The converse is false.

Indexing, transactions, concurrency: Two-phase lock = Growing phase, then shrinking phase. Strict 2PL holds exclusive locks until commit. 2PL guarantees conflict serializability. It can still deadlock.

08:00-10:00 2h THEORY**Normal forms, index, transaction**

Anomaly, functional dependency. 1NF, 2NF, 3NF, BCNF - example se. B-tree vs B+ tree, primary vs secondary index. ACID. Conflict serializability, precedence graph. Shared/exclusive lock, 2PL.

10:00-10:20 20 min BREAK**Break**

Test ka set abhi band rakho.

10:20-11:30 1h 10m PRACTICE**25 MCQ - normal form aur transaction**

Form identify karo. Precedence graph wala ek sawal khud banao.

25 technical MCQ**11:30-14:00 2h 30m BREAK****Khana aur aaram**

Sectional 2 baja hai. Subah wala topic doosri baar mat padho.

14:00-15:00 1h MOCK**Sectional: 50 sawal, 60 minute**

Algorithms + OS + DBMS. Timer. Har galat par -1 maan kar, isliye andaza kam. Paper-II ki practice yahin se hai.

50 technical MCQ**15:00-15:50 50 min ERROR LOG****Analysis**

Topic-wise galat count. Hafta 4 ki revision list mein teeno sabse kamzor topic likho.

15:50-16:30 40 min REASONING**Figure classification**

Embedded figure aur classification, 20 sawal.

20 reasoning**16:30-17:15 45 min GA****Static mix**

Polity, itihaas, economy se 20 sawal.

20 GA

18 Oct Ravivar

Normalization, transactions, hafta test (contd)

Padhai 6h 55m | 75 technical 20 reasoning 20 GA

17:15-17:45 30 min ERROR LOG

Agle hafte ki ek line

Kal Networks. OSI ke 7 naam raat ko yaad karne ki koshish mat karo - subah ka pehla kaam yahi hai.

19 Oct Somvar**Networks: layers aur data link**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Computer Networks: Concept of layering: OSI and TCP/IP protocol stacks

Paper-I + II 14.3.4 Computer Networks: Basics of packet, circuit and virtual circuit-switching

Paper-I + II 14.3.4 Computer Networks: Data link layer: framing, error detection, medium access control, Ethernet bridging

Paper-I 14.2.1 General Intelligence & Reasoning: Observation

Formula sheet for this day

Delay, framing, error, MAC: Four delays = Transmission = L/R . Propagation = d/v . Total = transmission + propagation + queue + processing. Bandwidth is R in bits per second. L is in bits. Do not mix bytes.

Delay, framing, error, MAC: Stop-and-wait = $a = T_p / T_t$. Utilization = $1 / (1 + 2a)$. $2a$ is the round trip measured in transmission times. A large a wastes the link.

Delay, framing, error, MAC: Sliding window = If $W < 1+2a$: utilization = $W/(1+2a)$. Else utilization = 1. W is the window in packets.

Sequence space must be at least $W+1$ for go-back- N in the simple model, and $2W$ for selective repeat.

Delay, framing, error, MAC: Nyquist aur Shannon = Noiseless Nyquist: $C = 2 * B * \log_2(M)$. Noisy Shannon: $C = B * \log_2(1 + S/N)$. If SNR is in dB, ratio = $10^{(dB/10)}$. M is the number of levels.

Delay, framing, error, MAC: Hamming = Distance d detects up to $d-1$ errors and corrects $\text{floor}((d-1)/2)$ errors. Parity adds distance. A single parity bit detects 1 and corrects 0.

Delay, framing, error, MAC: CRC = Treat the frame as a polynomial. Append remainder of division by the generator. A zero remainder at the receiver means no detected error. CRC is strong detection, not correction.

06:00-07:20 1h 20m THEORY**Layering aur switching**

OSI ke 7 aur TCP/IP ke layers, PDU ke naam. Circuit, packet, virtual circuit switching. Kaun si layer kya karti hai, device kis layer ka hai.

07:20-07:35 15 min BREAK**Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE**15 MCQ - layers**

Protocol ko layer se match karo. Galat match ek table mein.

15 technical MCQ**08:50-10:00 1h 10m BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE**Office, subah**

Kaam. 13:00 par Delay, framing, error, MAC ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK**Lunch: Delay, framing, error, MAC**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Delay, framing, error, MAC.

6 technical MCQ**13:20-16:30 3h 10m OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK**Chai: formula band karke**

12 minute, zaroori. Sheet band. Delay, framing, error, MAC mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

19 Oct Somvar**Networks: layers aur data link (contd)**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

16:42-18:50**2h 8m****OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00**10 min****OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Delay, framing, error, MAC.

19:00-20:00**1h****BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25**25 min****GA****Jagahon ke facts**

Straits, dams, capitals jo exams mein ghoomte hain. 15 facts likh lo.

10 GA**20:25-21:40****1h 15m****PRACTICE****Framing, CRC, Ethernet**

Parity, checksum, CRC numerical kam se kam 4. MAC, CSMA/CD ka idea, Ethernet, bridge vs router. Phir 25 MCQ. CRC ke steps error copy mein.

25 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Analogy, naya set**

20 sawal. Word aur number dono.

20 reasoning**22:35-22:50****15 min****ERROR LOG****CRC ke chaar step**

Generator polynomial se remainder tak, apne shabdon mein. 23:00 band.

20 Oct Mangalvar

IP aur subnetting

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Computer Networks: Routing protocols: shortest path, flooding, distance vector and link state routing

Paper-I + II 14.3.4 Computer Networks: Fragmentation and IP addressing, IPv4, CIDR notation

Paper-I + II 14.3.4 Computer Networks: Basics of IP support protocols (ARP, DHCP, ICMP) and NAT

Paper-I 14.2.1 General Intelligence & Reasoning: Relationship concepts

Formula sheet for this day

Routing, IPv4, CIDR, NAT: Routing idea = Distance vector: Bellman-Ford, advertise the whole vector, count-to-infinity. Link state: flood links, then Dijkstra at each router. Split horizon and poison reverse cut some, not all, loops.

Routing, IPv4, CIDR, NAT: IPv4 subnet = Prefix /p uses p network bits. Host bits = 32 - p. Usable hosts = $2^{(32-p)} - 2$. The -2 drops network and broadcast. /31 and /32 are special and do not follow this usable-host rule.

Routing, IPv4, CIDR, NAT: Mask from prefix = /p mask: p ones, then zeros. Block size of the interesting octet = $256 / 2^{(\text{ones in that octet})}$. A /26 interesting octet is the 4th: 2 bits host in that octet, block 64.

Routing, IPv4, CIDR, NAT: Fragment offset = Offset field is in 8-byte units. Byte offset = offset * 8. MF = 1 means more fragments follow. All fragments of a datagram carry the same identification. IPv4 header is 20 bytes minimum, 60 maximum. Total length includes the header.

Routing, IPv4, CIDR, NAT: Private and NAT = Private: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. NAT rewrites the private source to a public address, remembered in a table. ARP: IP to MAC on the local link. DHCP gives the IP. ICMP carries errors such as unreachable and echo.

06:00-07:20

1h 20m

THEORY

Routing

Flooding, distance vector, count-to-infinity, split horizon ka naam. Link state. Shortest path as a routing idea, algorithm kal wale graph se jod lo.

07:20-07:35

15 min

BREAK

Chhoti break

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50

1h 15m

PRACTICE

15 MCQ - routing

DV vs LS. Infinite count wala classic sawal.

15 technical MCQ

08:50-10:00

1h 10m

BREAK

Office ke liye nikalna

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00

3h

OFFICE

Office, subah

Kaam. 13:00 par Routing, IPv4, CIDR, NAT ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20

20 min

OFFICE DESK

Lunch: Routing, IPv4, CIDR, NAT

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Routing, IPv4, CIDR, NAT.

6 technical MCQ

13:20-16:30

3h 10m

OFFICE

Office, dopahar

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42

12 min

OFFICE DESK

Chai: formula band karke

12 minute, zaroori. Sheet band. Routing, IPv4, CIDR, NAT mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

20 Oct Mangalvar

IP aur subnetting (contd)

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

16:42-18:50 2h 8m OFFICE

Office, shaam

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 10 min OFFICE DESK

Nikalte hue 5 formulas

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Routing, IPv4, CIDR, NAT.

19:00-20:00 1h BREAK

Ghar aur khana

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25 25 min GA

Yojana aur mantralay

Das schemes, har ek ka mantralay ya ek line ka uddeshya.

10 GA

20:25-21:40 1h 15m PRACTICE

Subnetting drill

Classful limits, CIDR, network aur broadcast, usable hosts. Kam se kam 8 subnet khud se: /26, /27, /28, /29. ARP, DHCP, ICMP, NAT, fragmentation (offset, MF, DF). Phir 25 MCQ, inmein se adhe numerical.

25 technical MCQ

21:40-21:50 10 min BREAK

Das minute

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35 45 min REASONING

Coding, naya pattern

20 sawal. Symbol coding alag se 5.

20 reasoning

22:35-22:50 15 min ERROR LOG

Do subnet bina notes

Ek /26 aur ek /28. Host range likho. 23:00 band.

21 Oct Budhvar

Transport aur application

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Computer Networks: Transport layer: flow control, congestion control, UDP, TCP, sockets

Paper-I + II 14.3.4 Computer Networks: Application layer protocols: DNS, SMTP, HTTP, FTP, Email

Paper-I 14.2.1 General Intelligence & Reasoning: Arithmetical reasoning

Formula sheet for this day

TCP, UDP, application ports: TCP control = Sequence number counts bytes. Ack is the next byte expected. Window is the receiver's free buffer. UDP has no connection, no retransmission, no congestion window. Socket is (IP, port) at one end. A TCP connection is a pair of sockets.

TCP, UDP, application ports: Congestion window = Slow start: cwnd doubles each RTT until ssthresh. Then congestion avoidance: cwnd += 1 MSS per RTT. Timeout: ssthresh = cwnd/2, cwnd = 1 MSS. Reno triple duplicate ACK: fast retransmit, ssthresh = cwnd/2, then fast recovery. Tahoe sets cwnd to 1 on any loss signal. Reno keeps the pipe fuller on triple ACK.

TCP, UDP, application ports: Ports = DNS 53, DHCP 67/68, HTTP 80, HTTPS 443, SMTP 25, FTP 20 data and 21 control. DNS is usually UDP, TCP if the response is large. SMTP sends mail. HTTP fetches the web.

TCP, UDP, application ports: Flow vs congestion = Flow control: receiver window, do not overrun the receiver. Congestion control: cwnd, do not overrun the network. Send window = min(receiver window, cwnd). Silly window happens when the receiver opens a tiny window. Nagle and delayed ACK interact.

06:00-07:20 1h 20m THEORY

UDP, TCP, congestion

UDP vs TCP header ka farq, ports. Three-way handshake, FIN. Flow control vs congestion. Slow start ka idea. Socket kya hai, code nahi.

07:20-07:35 15 min BREAK

Chhoti break

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE

20 MCQ - transport

Handshake ke steps wale sawal. Galat sequence likho.

20 technical MCQ**08:50-10:00 1h 10m BREAK**

Office ke liye nikalna

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE

Office, subah

Kaam. 13:00 par TCP, UDP, application ports ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK

Lunch: TCP, UDP, application ports

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. TCP, UDP, application ports.

6 technical MCQ**13:20-16:30 3h 10m OFFICE**

Office, dopahar

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK

Chai: formula band karke

12 minute, zaroori. Sheet band. TCP, UDP, application ports mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

21 Oct Budhvar**Transport aur application (contd)**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

16:42-18:50**2h 8m****OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00**10 min****OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. TCP, UDP, application ports.

19:00-20:00**1h****BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25**25 min****GA****Science GK, chhota**

ISRO ke recent headings jo aapki current list mein hon, aur rozmarra computer full forms jo GA poochhta hai. Technical CN nahi.

10 GA**20:25-21:40****1h 15m****PRACTICE****DNS, mail, HTTP, mix CN**

DNS hierarchy, SMTP, HTTP, FTP, email ka flow (user agent se server tak). Phir 20 MCQ poore networks se: layer, CRC, subnet, TCP.

20 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Statement aur conclusion**

20 sawal. Conclusion ko statement se bahar mat le jao.

20 reasoning**22:35-22:50****15 min****ERROR LOG****TCP vs UDP, chhe point**

Connection, reliability, order, header, use-case. 23:00 band.

22 Oct Guruvar

Theory of Computation

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Theory of Computation: Regular expressions and finite automata

Paper-I + II 14.3.4 Theory of Computation: Context-free grammars and push-down automata

Paper-I + II 14.3.4 Theory of Computation: Regular and context-free languages, pumping lemma

Paper-I + II 14.3.4 Theory of Computation: Turing machines and undecidability

Paper-I 14.2.1 General Intelligence & Reasoning: Verbal classification

Formula sheet for this day

Theory of Computation: DFA aur NFA = DFA: $(Q, \Sigma, \delta, q_0, F)$, $\delta: Q \times \Sigma \rightarrow Q$, exactly one move. NFA: δ returns a set. Subset construction builds the DFA. Epsilon-NFA allows an empty move. Regular expressions, NFA, and DFA describe the same languages.

Theory of Computation: Pumping, regular = Exists p . For every w in L with $|w| \geq p$: $w = xyz$, $|xy| \leq p$, $|y| \geq 1$, and xy^kz is in L for every $k \geq 0$. Use it to prove a language is not regular. It cannot prove that a language is regular.

Theory of Computation: CFG aur PDA = CFG: (V, Σ, R, S) . Every context-free language has a PDA. Pumping for CFL: $w = uvxyz$, $|vxy| \leq p$, $|vy| \geq 1$, and uv^kxy^kz stays in L . A grammar is ambiguous if some string has two leftmost derivations or two parse trees.

Theory of Computation: TM aur decidability = Decidable: a TM halts with yes or no on every input. Recognizable: it halts on yes, and may loop on no. Halting problem is recognizable and undecidable. Type 3 regular, type 2 context-free, type 1 context-sensitive, type 0 unrestricted.

06:00-07:20 1h 20m THEORY

Regex, DFA, NFA

Language, regex, DFA vs NFA, epsilon-NFA. Conversion ka idea, poor algorithm ratna nahi. Chhoti language ka DFA bana lena.

07:20-07:35 15 min BREAK

Chhoti break

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE

20 MCQ - automata

Kaun si language regular hai. State count wale chhote sawal.

20 technical MCQ**08:50-10:00 1h 10m BREAK**

Office ke liye nikalna

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE

Office, subah

Kaam. 13:00 par Theory of Computation ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK

Lunch: Theory of Computation

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Theory of Computation.

6 technical MCQ**13:20-16:30 3h 10m OFFICE**

Office, dopahar

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK

Chai: formula band karke

12 minute, zaroori. Sheet band. Theory of Computation mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

22 Oct Guruvar**Theory of Computation (contd)**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

16:42-18:50**2h 8m****OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00**10 min****OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Theory of Computation.

19:00-20:00**1h****BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25**25 min****GA****Sanvaidhanik nikay**

Election Commission, CAG, UPSC, Finance Commission. Kaun banata hai, kya karta hai. Ek page.

10 GA**20:25-21:40****1h 15m****THEORY****CFG, PDA, TM, decidability**

CFG, leftmost derivation, PDA ka idea. Pumping lemma se language ko regular/CFL se bahar pehchanna, lamba proof nahi. TM. Halting problem undecidable hai - yeh list banao: decidable vs undecidable ke 6 classic naam. 20 MCQ.

20 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Number series**

20 sawal. Missing number aur galat number dono.

20 reasoning**22:35-22:50****15 min****ERROR LOG****Char class, ek example**

Regular, CFL, recursive, RE. Har ek ki ek language. 23:00 band.

23 Oct Shukravar

Compiler Design

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

Official syllabus lines

Paper-I + II 14.3.4 Compiler Design: Lexical analysis, parsing, syntax-directed translation

Paper-I + II 14.3.4 Compiler Design: Runtime environments

Paper-I + II 14.3.4 Compiler Design: Intermediate code generation

Paper-I + II 14.3.4 Compiler Design: Local optimization

Paper-I + II 14.3.4 Compiler Design: Data flow analysis: constant propagation, liveness analysis, common subexpression elimination

Paper-I 14.2.1 General Intelligence & Reasoning: Figure classification

Formula sheet for this day

Compiler Design: Phases = Lex $\rightarrow$ parse $\rightarrow$ semantic $\rightarrow$ intermediate code $\rightarrow$ optimize $\rightarrow$ target code. The symbol table is used across phases. Lexemes become tokens. Parse builds the tree. Semantic checks types.

Compiler Design: FIRST = FIRST(X) holds terminals that can begin a string derived from X. If X derives epsilon, epsilon is in FIRST(X). FIRST of a sequence: take FIRST of each symbol until one does not derive epsilon.

Compiler Design: FOLLOW = FOLLOW(A) holds terminals that can sit just after A in some sentential form. \$ is in FOLLOW of the start symbol. If $A \rightarrow \alpha B \beta$ and beta derives epsilon, FOLLOW(A) joins FOLLOW(B).

Compiler Design: LL(1) test = For $A \rightarrow \alpha \mid \beta$: FIRST(α) and FIRST(β) are disjoint. If α derives epsilon, FIRST(β) and FOLLOW(A) are disjoint. One lookahead. A conflict means the grammar is not LL(1), even if the language has some other LL(1) grammar.

Compiler Design: Shift-reduce = Actions: shift, reduce, accept, error. Shift/reduce and reduce/reduce are conflicts. Bottom-up. The handle is the right-hand side that is reduced.

Compiler Design: Local dataflow = Constant propagation: every reaching definition assigns the same constant. Liveness: a value is used on some path before the next definition. CSE: the same expression is computed again and operands are unchanged.

Three-address code has at most one operator: $x = y \text{ op } z$.

06:00-07:20**1h 20m****THEORY**

Phases aur tokens

Compiler ki phases ka order. Lexeme vs token. Symbol table kab banti hai. Parse tree vs syntax tree. Error kis phase mein pakda jaata hai.

07:20-07:35**15 min****BREAK**

Chhoti break

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50**1h 15m****PRACTICE**

20 MCQ - phases

Kaun sa kaam kis phase ka hai. Galat phase ko sahi ke saath likho.

20 technical MCQ**08:50-10:00****1h 10m****BREAK**

Office ke liye nikalna

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00**3h****OFFICE**

Office, subah

Kaam. 13:00 par Compiler Design ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20**20 min****OFFICE DESK**

Lunch: Compiler Design

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Compiler Design.

6 technical MCQ

23 Oct Shukravar**Compiler Design (contd)**

Padhai 5h 57m | 46 technical 20 reasoning 10 GA

13:20-16:30**3h 10m****OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42**12 min****OFFICE DESK****Chai: formula band karke**

12 minute, zaroori. Sheet band. Compiler Design mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50**2h 8m****OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00**10 min****OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Compiler Design.

19:00-20:00**1h****BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25**25 min****GA****Polity, ek page revise**

Rights, DPSP, parliament, emergency. 10 rapid sawal apni purani sheet se.

10 GA**20:25-21:40****1h 15m****THEORY****FIRST, FOLLOW, optimization**

Ek chhota grammar: FIRST aur FOLLOW. LL(1) conflict ka matlab. Shift-reduce, handle. Three-address code ka ek example.

Constant folding, constant propagation, liveness, common subexpression - har ek ka ek line example. 20 MCQ.

20 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Direction aur blood, mix**

20 sawal, aadhe-aadhe. Diagram compulsory.

20 reasoning**22:35-22:50****15 min****ERROR LOG****FIRST / FOLLOW ek grammar**

Bina notes ke wahi chhota grammar dubara. 23:00 band.

24 Oct Shanivar

Maths: discrete aur linear algebra

Padhai 8h 45m | 50 technical 20 reasoning 15 GA

Official syllabus lines

Paper-I + II 14.3.4 Engineering Mathematics: Discrete Mathematics: propositional and first-order logic
 Paper-I + II 14.3.4 Engineering Mathematics: Sets, relations, functions
 Paper-I + II 14.3.4 Engineering Mathematics: Partial orders and lattices
 Paper-I + II 14.3.4 Engineering Mathematics: Monoids, groups
 Paper-I + II 14.3.4 Engineering Mathematics: Graphs: connectivity, matching, colouring
 Paper-I + II 14.3.4 Engineering Mathematics: Combinatorics: counting, recurrence relations, generating functions
 Paper-I + II 14.3.4 Engineering Mathematics: Linear algebra: matrices, determinants, systems of linear equations
 Paper-I + II 14.3.4 Engineering Mathematics: Eigenvalues and eigenvectors, LU decomposition
 Paper-I 14.2.1 General Intelligence & Reasoning: Arithmetical number series

Formula sheet for this day

Discrete mathematics: Implication = $p \rightarrow q$ is equivalent to $(\text{not } p) \text{ or } q$. Contrapositive $\text{not } q \rightarrow \text{not } p$ is equivalent. Converse $q \rightarrow p$ is not. Inverse is $\text{not } p \rightarrow \text{not } q$. It matches the converse, not the original.

Discrete mathematics: Set sizes = $|A \cup B| = |A| + |B| - |A \cap B|$. De Morgan: $(A \cup B)' = A' \cap B'$. Three sets: add singles, subtract pairs, add back the triple intersection.

Discrete mathematics: Relations = Equivalence: reflexive, symmetric, transitive. Partial order: reflexive, antisymmetric, transitive. A partial order is a poset. A lattice gives every pair a join and a meet.

Discrete mathematics: Group = Closure, associativity, identity, inverse. Abelian if also commutative. Lagrange: subgroup order divides $|G|$. A monoid has closure, associativity, identity, and may lack inverses.

Discrete mathematics: Graph counts = Sum of degrees = $2|E|$. Tree: connected, $|E| = |V| - 1$. Complete graph edges = $n(n-1)/2$. Bipartite iff no odd cycle. Handshaking says the number of odd-degree vertices is even.

Discrete mathematics: Counting = $nPr = n! / (n-r)!$. $nCr = n! / (r! (n-r)!)$. Sum of geometric series $a + ar + \dots + ar^{(n-1)} = a(r^n - 1) / (r - 1)$, $r \neq 1$. Recurrence: solve the characteristic equation for linear constant coefficients. Generating function $A(x) = \sum a_n x^n$.

Linear algebra: Determinant = $\det(AB) = \det(A) \det(B)$. $\det(A^T) = \det(A)$. $\det(cA) = c^n \det(A)$ for an n by n matrix. Inverse exists iff $\det \neq 0$. $(AB)^{-1} = B^{-1} A^{-1}$.

Linear algebra: Eigen = $A v = \lambda v$, $v \neq 0$. Characteristic equation $\det(A - \lambda I) = 0$. Trace = sum of eigenvalues. $\det$ = product of eigenvalues. A symmetric real matrix has real eigenvalues and orthogonal eigenvectors.

Linear algebra: $LU = A = L U$. Solve $L y = b$ by forward substitution, then $U x = y$ by back substitution. $P A = L U$ when a pivot swap is required. L is unit lower triangular in the usual form.

Linear algebra: Linear system = Unique solution if $\text{rank}(A) = \text{rank}([A|b]) = n$. Infinite if $\text{rank}(A) = \text{rank}([A|b]) < n$. None if $\text{rank}(A) < \text{rank}([A|b])$. Rank is the number of nonzero rows in row-echelon form. Rank-nullity: $\text{rank} + \text{nullity} = \text{number of columns}$.

07:00-09:00**2h****THEORY****Discrete mathematics**

Propositional logic, sets, relations: reflexive, symmetric, transitive, equivalence. Partial order, Hasse diagram, lattice. Functions. Groups ke axioms, example level. Counting, permutation, combination. Recurrence ka simple form. Graph words: connectivity, matching, colouring.

09:00-09:30**30 min****BREAK****Nashta**

Aadha ghanta.

09:30-11:15**1h 45m****PRACTICE****25 MCQ - discrete**

Relation properties wale sawal sabse pehle. Galat property ko example ke saath likho.

25 technical MCQ**11:15-12:00****45 min****REASONING****Ek puzzle set**

20 minute setup, phir sawal. 20 se zyada mat badhao agar set lamba ho.

20 reasoning

24 Oct Shanivar

Maths: discrete aur linear algebra (contd)

Padhai 8h 45m | 50 technical 20 reasoning 15 GA

12:00-14:30 2h 30m BREAK

Khana aur aaram

Matrices 14:30 se.

14:30-16:15 1h 45m THEORY

Linear algebra

Matrix operations, determinant, rank, linear system kab solution hai. Eigenvalue aur eigenvector 2x2 par haath se. LU decomposition sirf yeh ki kya toda jaata hai.

16:15-16:45 30 min GA

Economy, current

Apni 10 headings mein se economy wale. 15 facts ya sawal.

15 GA

16:45-18:15 1h 30m PRACTICE

25 MCQ - matrices

2x2 eigen haath se. Rank wale sawal. Calculator nahi.

25 technical MCQ

18:15-18:45 30 min ERROR LOG

Formula ki aadhi sheet

Relation properties aur det/eigen. Kal probability ki sheet isi mein judegi. Aaj band.

25 Oct Ravivar**Calculus, probability, hafta test**

Padhai 7h 20m | 75 technical 25 reasoning 20 GA

Official syllabus lines

Paper-I + II 14.3.4 Engineering Mathematics: Calculus: limits, continuity and differentiability

Paper-I + II 14.3.4 Engineering Mathematics: Maxima and minima, mean value theorem, integration

Paper-I + II 14.3.4 Engineering Mathematics: Random variables. Uniform, normal, exponential, Poisson and binomial distributions

Paper-I + II 14.3.4 Engineering Mathematics: Mean, median, mode and standard deviation

Paper-I + II 14.3.4 Engineering Mathematics: Conditional probability and Bayes' theorem

Paper-I 14.2.1 General Intelligence & Reasoning: Abstract ideas and symbols and their relationships

Formula sheet for this day

Calculus: Limits to remember = $\sin x / x \rightarrow 1$ as $x \rightarrow 0$. $(1+x)^{1/x} \rightarrow e$ as $x \rightarrow 0$. $(1+1/n)^n \rightarrow e$. Continuity at a: limit equals the function value. Differentiable implies continuous. The converse is false.

Calculus: Derivatives = $(uv)' = u'v + uv'$. $(u/v)' = (u'v - uv') / v^2$. Chain: $dy/dx = dy/du * du/dx$. $x^n \rightarrow n x^{n-1}$. $e^x \rightarrow e^x$. $\ln x \rightarrow 1/x$. $\sin \rightarrow \cos$. $\cos \rightarrow -\sin$. A critical point has $f' = 0$ or f' undefined.

Calculus: MVT aur extrema = Rolle: $f(a)=f(b)$ implies some c with $f'(c)=0$. MVT: $f'(c) = (f(b)-f(a))/(b-a)$. Local max: $f'=0$ and $f''<0$. Local min: $f'=0$ and $f''>0$. If $f''=0$ the test fails. Check the sign of f' around the point.

Calculus: Integrals = $\int x^n dx = x^{n+1}/(n+1) + C$, $n \neq -1$. $\int dx/x = \ln|x| + C$. $\int e^x dx = e^x + C$. Parts: $\int u dv = uv - \int v du$. Definite integral is net area. Swap limits and the sign flips.

Probability and statistics: Conditional aur Bayes = $P(A|B) = P(A \text{ and } B) / P(B)$. Bayes: $P(A|B) = P(B|A) P(A) / P(B)$. Total: $P(B) = \sum P(B|A_i) P(A_i)$. Independent means $P(A \text{ and } B) = P(A)P(B)$, so $P(A|B)=P(A)$.

Probability and statistics: Mean aur spread = $E[X] = \sum x p(x)$. $\text{Var} = E[X^2] - (E[X])^2$. $SD = \sqrt{\text{Var}}$. Median splits the ordered list. Mode is the most frequent value. For grouped data the mean uses class marks. Two modes: bimodal.

Probability and statistics: Binomial = $P(K=k) = C(n,k) p^k (1-p)^{n-k}$. Mean = np . Variance = $np(1-p)$. n independent trials, two outcomes, constant p .

Probability and statistics: Poisson = $P(K=k) = e^{-\lambda} * \lambda^k / k!$. Mean = variance = λ . Use for rare events in a fixed interval. It is the limit of binomial when n is large and p is small, $\lambda = np$.

Probability and statistics: Uniform, exponential, normal = Uniform $[a,b]$: mean $(a+b)/2$, variance $(b-a)^2 / 12$. Exponential: mean $1/\lambda$, variance $1/\lambda^2$, memoryless. Normal $z = (x - \mu) / \sigma$. Exponential pdf is $\lambda e^{-\lambda x}$ for $x \geq 0$.

Standard normal has $\mu 0$ and $\sigma 1$.

08:00-09:20**1h 20m****THEORY****Calculus, MCQ jitna**

Limits, continuity, differentiability. Maxima-minima first derivative se. Mean value theorem ka statement. Definite integration ke simple sawal. Lamba technique nahi.

09:20-09:30**10 min****BREAK****Das minute**

Probability ka formula page khali rakho.

09:30-10:50**1h 20m****THEORY****Probability aur distributions**

Conditional probability, Bayes. Mean, median, mode, standard deviation. Binomial, Poisson, uniform, exponential, normal - sirf mean aur variance ke formula, aur kab kaun si distribution.

10:50-11:40**50 min****PRACTICE****25 MCQ - calculus aur probability**

Bayes ke do poore sawal haath se. Baaki formula identify.

25 technical MCQ**11:40-14:00****2h 20m****BREAK****Khana aur aaram**

Hafte ka aakhri test 2 baje. Formula sheet saath rakhni hai, kholni nahi.

14:00-15:00**1h****MOCK****Sectional: 50 sawal, 60 minute**

Networks + TOC + Compiler + Maths. -1 maan kar. Subnet aur Bayes aayein to pehle wahi, kyunki wahan number hai.

50 technical MCQ

25 Oct Ravivar

Calculus, probability, hafta test (contd)

Padhai 7h 20m | 75 technical 25 reasoning 20 GA

15:00-15:50 50 min ERROR LOG

Analysis

Kaun se topic 50% se neeche hain. Wahi 26-29 Oct ki revision mein pehle aayenge.

15:50-16:35 45 min REASONING

Non-verbal sectional

25 sawal, mix figure. Timer 25 minute.

25 reasoning

16:35-17:20 45 min GA

Static 20

Polity aur itihaas se zyada, kyunki wahan dohraane layak cheez hai.

20 GA

17:20-17:50 30 min ERROR LOG

Distribution sheet band

Mean/variance ki chhoti table. Kal se naya chapter nahi. Sirf revision.

26 Oct Somvar

Revision: Logic, COA, DS

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

Aaj se koi naya chapter nahi. Notes sirf formula sheet jitne khulenge.

Official syllabus lines

Paper-I 14.2.1 General Intelligence & Reasoning: Arithmetical computations

Paper-I + II 14.3.4 Revision: Second pass: Digital Logic, Computer Organization, Programming and Data Structures

Formula sheet for this day

Digital Logic: De Morgan = $(A+B)' = A'B'$ $(AB)' = A'+B'$. NAND is OR of complements. NOR is AND of complements.Digital Logic: XOR aur XNOR = $A \oplus B = A'B + AB'$ $XNOR = AB + A'B'$. XOR is 1 when bits differ. XNOR is 1 when bits match.Digital Logic: Absorption = $A + AB = A$ $A(A+B) = A$ $A + A'B = A + B$. Drop the larger product term when one term contains the other.Digital Logic: 2's complement = Invert every bit, then add 1. Range of n bits: $-2^{(n-1)} \dots 2^{(n-1)}-1$. Unsigned range is $0 \dots 2^n - 1$.

One's complement is invert only.

Digital Logic: r's complement = $(r-1)$'s complement = $(r^n - 1) - N$. r's complement = $r^n - N$. For decimal, 9's then add 1 gives 10's complement.Digital Logic: Flip-flop next state = SR (SR=0): $Q+ = S + R'Q$. JK: $Q+ = JQ' + K'Q$. D: $Q+ = D$. T: $Q+ = T \oplus Q$. SR = 11 is invalid. JK = 11 toggles. T = 1 toggles.Digital Logic: Counter aur MUX = n flip-flops, ripple up-counter: $\text{mod } 2^n$, $f_{\text{out}} = f_{\text{in}} / 2^n$. MUX: 2^n data, n select. Decoder: n to 2^n . A mod-10 counter needs extra reset logic. It is not a pure 4-bit free counter.

Digital Logic: K-map group = Group size must be 1, 2, 4, 8, ... Circle power-of-two 1s (or 0s for POS). Don't-care may join a group, never a group of its own. Adjacent cells differ by one bit, including wrap of the edges.

Digital Logic: IEEE-754 single = 1 sign, 8 exponent, 23 fraction. Bias 127. Normal value = $(-1)^s * (1.f) * 2^{(E-127)}$. E = 0 and f = 0 is zero. E = 255 and f = 0 is infinity. E = 255 and f != 0 is NaN. Double bias is 1023.Computer Organization: CPU time = CPU time = IC * CPI * clock cycle. Clock cycle = $1 / \text{frequency}$. IC is instruction count. A lower CPI or a faster clock both cut time.Computer Organization: Pipeline speedup = No stalls, n instructions, k stages: $S = (n*k) / (k + n - 1)$. Large n: S approaches k. One stall cycle drops the ideal throughput. Efficiency = S / k .Computer Organization: Stall CPI = Ideal pipeline CPI = 1. Effective CPI = $1 + \text{stall cycles per instruction}$. Data hazard: forwarding removes many, not all, stalls. Control hazard: branch delay or flush.Computer Organization: Cache AMAT = AMAT = Hit time + Miss rate * Miss penalty. Miss rate = $1 - \text{hit rate}$. Two-level: $AMAT = T_{L1} + M1*(T_{L2} + M2*T_{\text{mem}})$.Computer Organization: Hit-rate EAT = EAT = $h*T_{\text{cache}} + (1-h)*T_{\text{memory}}$. If every access also pays the cache lookup: $EAT = h*T_c + (1-h)*(T_c+T_m)$. Use the form the question states.Computer Organization: Cache address split = Offset = $\log_2(\text{block bytes})$. Direct index = $\log_2(\text{blocks})$. Set index = $\log_2(\text{sets})$. Tag = address bits - index - offset. Sets = blocks / ways. Cache bytes = sets * ways * block bytes.

Computer Organization: Effective address = Direct: EA = address field. Indirect: EA = M[address]. Indexed: EA = index + address. Relative: EA = PC + offset. Register-indirect: EA = register. Immediate has no memory EA. The operand sits in the instruction.

Computer Organization: Interrupt vs DMA = Interrupt: CPU saves context and runs the handler. DMA: device moves a block; CPU is told at the end. Cycle stealing takes the bus for single transfers. Burst mode holds the bus for the block.

C, arrays, stacks, queues, lists: Array address = 1D: base + i * size. Row-major $A[i][j] = \text{base} + (i*\text{cols} + j)*\text{size}$. Column-major: base + $(j*\text{rows} + i)*\text{size}$. C uses row-major and 0-based indexes.C, arrays, stacks, queues, lists: Pointer step = p + k moves k elements, not k bytes. Byte move = $k * \text{sizeof}(*p)$. $a[i]$ is $*(a+i)$. $\&a[i]$ is $a+i$.C, arrays, stacks, queues, lists: Stack aur queue = Stack: push and pop at the same end, $O(1)$. Circular queue: front and rear mod capacity, $O(1)$. Infix to postfix uses a stack. BFS uses a queue. DFS uses a stack.C, arrays, stacks, queues, lists: Linked list = Insert or delete at a known node: $O(1)$. Search: $O(n)$. Extra memory: one pointer per node. Doubly linked delete of a known node does not need the previous walk.C, arrays, stacks, queues, lists: Recursion cost = Time follows the recurrence of the calls. Depth of the call stack is the deepest chain, often $O(n)$ or $O(\log n)$. A missing base case overflows the stack. Tail recursion can reuse the frame.Trees, heaps, graphs: Binary tree counts = Max nodes at level L (root level 0) = 2^L . Perfect tree height h: nodes = $2^{(h+1)} - 1$. Full tree: leaves = internal nodes + 1. Height of a single node is 0 in this sheet. Some books use 1. Match the question.Trees, heaps, graphs: BST = Left subtree < node < right subtree. Search, insert, delete: $O(h)$. Balanced h = $\text{floor}(\log_2 n)$. Skewed h = n-1. Inorder of a BST is sorted order.Trees, heaps, graphs: Binary heap index = 0-based: parent $\text{floor}((i-1)/2)$, left $2i+1$, right $2i+2$. Height $\text{floor}(\log_2 n)$. Build $O(n)$. Extract $O(\log n)$. Min-heap: parent <= children. Heap is a complete binary tree.Trees, heaps, graphs: Graph storage = Adjacency matrix: $O(V^2)$ space, edge check $O(1)$. Adjacency list: $O(V+E)$ space, edge check $O(\text{degree})$. Undirected edge appears twice in the list, once in the matrix triangle.

26 Oct Somvar**Revision: Logic, COA, DS (contd)**

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

06:00-07:20 **1h 20m** **ERROR LOG****Formula sheet: DL, COA, DS**

K-map rules, flip-flop equations, speedup, cache, complexities of list/stack/tree. Padhna nahi, yaad se likhna. Jo na aaye sirf wahi line notes mein dekho.

07:20-07:35 **15 min** **BREAK****Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 **1h 15m** **PRACTICE****30 MCQ mix**

Digital Logic, COA, DS. Timer rakho, 30 sawal 50 minute mein, baaki review.

30 technical MCQ**08:50-10:00** **1h 10m** **BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 **3h** **OFFICE****Office, subah**

Kaam. 13:00 par Digital Logic, Computer Organization, C, arrays, stacks, queues, lists, Trees, heaps, graphs ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 **20 min** **OFFICE DESK****Lunch: Digital Logic, Computer Organization, C, arrays, stacks, queues, lists, Trees, heaps, graphs**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Digital Logic, Computer Organization, C, arrays, stacks, queues, lists, Trees, heaps, graphs.

6 technical MCQ**13:20-16:30** **3h 10m** **OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 **12 min** **OFFICE DESK****Chai: formula band karke**

12 minute, zaroori. Sheet band. Digital Logic, Computer Organization, C, arrays, stacks, queues, lists, Trees, heaps, graphs mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 **2h 8m** **OFFICE****Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 **10 min** **OFFICE DESK****Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Digital Logic, Computer Organization, C, arrays, stacks, queues, lists, Trees, heaps, graphs.

19:00-20:00 **1h** **BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25 **25 min** **GA****Static rapid**

Purani sheets se 15 sawal. Naya fact tabhi jab purana galat hua ho.

15 GA

26 Oct Somvar**Revision: Logic, COA, DS (contd)**

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

20:25-21:40**1h 15m****PRACTICE****Hafta 1 ke galat + 30 naye**

Error copy ke Digital Logic/COA/DS sawal pehle. Phir 30 fresh MCQ. Jo doosri baar bhi galat ho, us par 10 minute concept, usse zyada nahi.

30 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Mix 25**

Analogy, coding, series, syllogism. Sectional jaisa, 25 minute try karo.

25 reasoning**22:35-22:50****15 min****ERROR LOG****Kal ke numerical mark karo**

OS scheduling aur page replacement kal hain. Formula sheet par unke naam likh do. 23:00 band.

27 Oct Mangalvar

Revision: Algorithms aur OS

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

Official syllabus lines

Paper-I 14.2.1 General Intelligence & Reasoning: Verbal and non-verbal questions

Paper-I + II 14.3.4 Revision: Second pass: Algorithms and Operating System

Formula sheet for this day

Searching, sorting, hashing: Binary search = Comparisons about $\text{floor}(\log_2 n) + 1$. Needs a sorted array. Time $\Theta(\log n)$. Lower bound of comparison sort is $\Omega(n \log n)$.

Searching, sorting, hashing: Sorting cost = Insertion, selection, bubble: worst $\Theta(n^2)$. Merge: $\Theta(n \log n)$, extra $O(n)$. Heap: $\Theta(n \log n)$, in place. Quick: average $\Theta(n \log n)$, worst $\Theta(n^2)$. Insertion best case $\Theta(n)$ on sorted input. Merge and heap are stable? Merge is stable. Heap and quick are not.

Searching, sorting, hashing: Stability = Stable: equal keys keep their input order. Merge and insertion are stable. Selection, heap, and quick are not. Counting sort is stable if the prefix scan fills from the end.

Searching, sorting, hashing: Counting aur radix = Counting: $O(n + k)$ time, k is the key range. Radix: $O(d(n + k))$ for d digits. Use when keys are integers in a known range, not for arbitrary comparisons.

Searching, sorting, hashing: Hash load = $\alpha = n / m$. Chaining, unsuccessful search about $1 + \alpha$. Successful about $1 + \alpha/2$. Open addressing degrades as α approaches 1. Keep α well below 1.

Greedy, DP, divide-and-conquer: Master theorem = $T(n) = a T(n/b) + f(n)$. Let $c = \log_b(a)$. If $f = O(n^{c-\epsilon})$: $\Theta(n^c)$. If $f = \Theta(n^c)$: $\Theta(n^c \log n)$. If $f = \Omega(n^{c+\epsilon})$ and $a f(n/b) \leq k f(n)$ for $k < 1$: $\Theta(f(n))$. ϵ is a positive constant. Merge sort: $a=2, b=2, f=n, c=1$, case 2.

Greedy, DP, divide-and-conquer: Divide and conquer = Split, solve, combine. Binary search, merge sort, quicksort, Strassen. Cost is the recurrence, then Master theorem. Not every split is equal. Quicksort's worst split is 1 and $n-1$.

Greedy, DP, divide-and-conquer: Greedy choice = Feasible choice that looks best now, plus optimal substructure. Fractional knapsack: sort by value/weight. 0/1 knapsack is not greedy. It is DP. Activity selection: pick the finish-earliest.

Greedy, DP, divide-and-conquer: DP shape = Optimal substructure and overlapping subproblems. 0/1 knapsack: $dp[i][w] = \max(dp[i-1][w], v_i + dp[i-1][w-w_i])$. Fibonacci naive recursion is exponential. With a table it is $O(n)$.

Graph algorithms: BFS aur DFS = Time $O(V+E)$ on a list. BFS gives fewest edges from the source in an unweighted graph. DFS gives finishing times. Topological sort: decreasing finish time, and only on a DAG.

Graph algorithms: MST = Kruskal: sort edges, add if no cycle, $O(E \log E)$. Prim: grow one tree, $O(E \log V)$ with a heap. MST has $V-1$ edges. Cut property: lightest edge across a cut is in some MST. Negative weights are allowed. Negative cycles are not a concept here.

Graph algorithms: Dijkstra = Non-negative weights. Heap version $O((V+E) \log V)$. $\text{Dist}[v] = \min(\text{dist}[v], \text{dist}[u] + w(u,v))$ when u is settled. A negative edge can make a settled node wrong. Use Bellman-Ford then.

Graph algorithms: Bellman-Ford aur Floyd = Bellman-Ford: relax all edges $V-1$ times, $O(VE)$. One more relax means a negative cycle. Floyd: $dp[k][i][j] = \min(dp[i][j], dp[i][k] + dp[k][j])$, $O(V^3)$. Floyd allows negative edges, not a negative cycle.

Processes, sync, deadlock: TAT, waiting, response = TAT = completion - arrival. Waiting = TAT - burst. Response = first CPU start - arrival. CPU utilization = busy time / total time. Throughput = jobs finished / time.

Processes, sync, deadlock: Scheduling pick = FCFS: simple, convoy. SJF: least average waiting if all arrive together, non-preemptive. SRTF: preemptive SJF. RR: quantum q , more switches as q shrinks. Priority can starve unless aging raises the waiting job.

Processes, sync, deadlock: Banker's need = $\text{Need}[i] = \text{Max}[i] - \text{Allocation}[i]$. Safe if a sequence exists: Work starts as Available, and each picked process has $\text{Need} \leq \text{Work}$, then $\text{Work} += \text{Allocation}$. An unsafe state is not a deadlock yet. It might become one.

Processes, sync, deadlock: Deadlock four = Mutual exclusion, hold and wait, no preemption, circular wait. All four together. Break any one condition and deadlock cannot form. Prevention vs avoidance vs detection.

Processes, sync, deadlock: Semaphore = Wait (P): while $S \leq 0$ loop; $S = S-1$. Signal (V): $S = S+1$. Binary semaphore is 0 or 1. Counting semaphore tracks a pool. Mutex lock is not a signal to a different process by itself. A semaphore can.

Memory and file systems: Page bits = Offset bits = $\log_2(\text{page size})$. Page number = logical address bits - offset bits. Frames = physical bytes / page size. Logical pages and physical frames are the same size.

Memory and file systems: TLB EAT = Single-level page table: $\text{EAT} = h*(T_{\text{tlb}} + T_{\text{mem}}) + (1-h)*(T_{\text{tlb}} + 2*T_{\text{mem}})$. A miss reads the page table in memory, then the data. Two memory reads.

Memory and file systems: Page fault time = $\text{EAT} = (1-p)*\text{memory access} + p*\text{page-fault service}$. p is the fault probability. Service includes disk. It dominates memory.

Memory and file systems: Replacement = FIFO can show Belady: more frames, more faults. Optimal: replace the page used farthest in the future. LRU: replace the least recently used. Optimal needs the future. It is a yardstick, not an online policy.

Memory and file systems: Fragmentation = Paging: internal fragment in the last page only. Segmentation: external fragment between variable segments. Compaction moves segments. Paging does not need it for external holes.

Memory and file systems: File allocation = Contiguous: direct, external fragment. Linked: no external fragment, slow random.

27 Oct Mangalvar**Revision: Algorithms aur OS (contd)**

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

Indexed: index block, random access, one extra read. Unix inode: direct blocks, then single, double, triple indirect.

06:00-07:20 1h 20m ERROR LOG**Sort aur schedule, yaad se**

8 sorts ki complexity. FCFS, SJF, SRTF, RR ke rules. FIFO, LRU, Optimal ka farq. Likh kar dekho.

07:20-07:35 15 min BREAK**Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE**Numerical + 20 MCQ**

4 scheduling Gantt aur 3 page-replacement. Phir 20 MCQ algorithms se. Kul milakar 30 sawal ke barabar kaam.

30 technical MCQ**08:50-10:00 1h 10m BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE**Office, subah**

Kaam. 13:00 par Searching, sorting, hashing, Greedy, DP, divide-and-conquer, Graph algorithms, Processes, sync, deadlock, Memory and file systems ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK**Lunch: Searching, sorting, hashing, Greedy, DP, divide-and-conquer, Graph algorithms, Processes, sync, deadlock, Memory and file systems**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Searching, sorting, hashing, Greedy, DP, divide-and-conquer, Graph algorithms, Processes, sync, deadlock, Memory and file systems.

6 technical MCQ**13:20-16:30 3h 10m OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK**Chai: formula band karke**

12 minute, zaroori. Sheet band. Searching, sorting, hashing, Greedy, DP, divide-and-conquer, Graph algorithms, Processes, sync, deadlock, Memory and file systems mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 2h 8m OFFICE**Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 10 min OFFICE DESK**Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Searching, sorting, hashing, Greedy, DP, divide-and-conquer, Graph algorithms, Processes, sync, deadlock, Memory and file systems.

19:00-20:00 1h BREAK**Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

27 Oct Mangalvar

Revision: Algorithms aur OS (contd)

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

20:00-20:25 25 min GA

Static rapid

Geo aur culture ki purani list, 15 sawal.

15 GA

20:25-21:40 1h 15m PRACTICE

Hafta 2 ki galtiyen + 30 MCQ

Algo aur OS ke marked wrong pehle. Phir 30 fresh. DP table wala ek sawal zaroor ho.

30 technical MCQ

21:40-21:50 10 min BREAK

Das minute

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35 45 min REASONING

Mix 25

Blood, direction, inequality, seating ka ek chhota set.

25 reasoning

22:35-22:50 15 min ERROR LOG

SQL aur subnet ka

Error copy se un dono ke galat alag kar do. 23:00 band.

28 Oct Budhvar**Revision: DBMS aur Networks**

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

Official syllabus lines

Paper-I 14.2.1 General Intelligence & Reasoning: Analytical functions

Paper-I + II 14.3.4 Revision: Second pass: Databases and Computer Networks

Formula sheet for this day

ER, algebra, SQL, normal forms: Relational algebra = sigma select, pi project, rho rename, union, minus, cartesian product, join. Select keeps rows. Project keeps columns and removes duplicate rows.

ER, algebra, SQL, normal forms: Keys = Superkey uniquely identifies. Candidate key is a minimal superkey. Primary key is one chosen candidate. Foreign key references a candidate key. Prime attribute: member of some candidate key.

ER, algebra, SQL, normal forms: Normal forms = 1NF: atomic values. 2NF: no partial dependency of a non-prime on a candidate key. 3NF: for $X \rightarrow A$, X is a superkey or A is prime. BCNF: for every nontrivial $X \rightarrow A$, X is a superkey. BCNF implies 3NF. 3NF does not imply BCNF. A table can be 3NF and still have overlapping keys.

ER, algebra, SQL, normal forms: Lossless join = Decomposition of two parts is lossless if $R_1 \cap R_2$ is a key of R_1 or of R_2 .

Dependency preserving: the union of the projected FDs is equivalent to F. Lossless and preserving are different tests.

ER, algebra, SQL, normal forms: SQL groups = WHERE filters rows before groups. HAVING filters groups. SELECT aggregates need GROUP BY of the other selected columns. COUNT(*) counts rows. COUNT(col) skips NULL. DISTINCT inside COUNT counts uniques.

Indexing, transactions, concurrency: B and B+ tree = Order-m node holds up to m children. Search $O(\log n)$. B+ tree stores records only in the leaves, and leaves are linked. A B+ leaf scan is the range query. Internal nodes are a sparse directory.

Indexing, transactions, concurrency: Index kind = Dense: one entry per record. Sparse: one entry per block, file must be ordered.

Primary index is on the ordering key. Secondary index is on another field. Clustering index can be on a non-unique ordered field.

Indexing, transactions, concurrency: ACID = Atomic: all or nothing. Consistent: constraints hold. Isolated: concurrent effect matches some serial order. Durable: commit survives a crash. A schedule can be atomic per transaction and still not serializable.

Indexing, transactions, concurrency: Serializability = Conflict: two ops, different transactions, same item, at least one write. Conflict serializable iff the precedence graph has no cycle. Conflict serializable implies view serializable. The converse is false.

Indexing, transactions, concurrency: Two-phase lock = Growing phase, then shrinking phase. Strict 2PL holds exclusive locks until commit. 2PL guarantees conflict serializability. It can still deadlock.

Delay, framing, error, MAC: Four delays = Transmission = L/R . Propagation = d/v . Total = transmission + propagation + queue + processing. Bandwidth is R in bits per second. L is in bits. Do not mix bytes.

Delay, framing, error, MAC: Stop-and-wait = $a = T_p / T_t$. Utilization = $1 / (1 + 2a)$. $2a$ is the round trip measured in transmission times. A large a wastes the link.

Delay, framing, error, MAC: Sliding window = If $W < 1 + 2a$: utilization = $W / (1 + 2a)$. Else utilization = 1. W is the window in packets.

Sequence space must be at least $W + 1$ for go-back-N in the simple model, and $2W$ for selective repeat.

Delay, framing, error, MAC: Nyquist aur Shannon = Noiseless Nyquist: $C = 2 * B * \log_2(M)$. Noisy Shannon: $C = B * \log_2(1 + S/N)$. If SNR is in dB, ratio = $10^{(dB/10)}$. M is the number of levels.

Delay, framing, error, MAC: Hamming = Distance d detects up to $d - 1$ errors and corrects $\lfloor (d - 1) / 2 \rfloor$ errors. Parity adds distance. A single parity bit detects 1 and corrects 0.

Delay, framing, error, MAC: CRC = Treat the frame as a polynomial. Append remainder of division by the generator. A zero remainder at the receiver means no detected error. CRC is strong detection, not correction.

Routing, IPv4, CIDR, NAT: Routing idea = Distance vector: Bellman-Ford, advertise the whole vector, count-to-infinity. Link state: flood links, then Dijkstra at each router. Split horizon and poison reverse cut some, not all, loops.

Routing, IPv4, CIDR, NAT: IPv4 subnet = Prefix /p uses p network bits. Host bits = $32 - p$. Usable hosts = $2^{(32-p)} - 2$. The -2 drops network and broadcast. /31 and /32 are special and do not follow this usable-host rule.

Routing, IPv4, CIDR, NAT: Mask from prefix = /p mask: p ones, then zeros. Block size of the interesting octet = $256 / 2^{\text{(ones in that octet)}}$. A /26 interesting octet is the 4th: 2 bits host in that octet, block 64.

Routing, IPv4, CIDR, NAT: Fragment offset = Offset field is in 8-byte units. Byte offset = offset * 8. MF = 1 means more fragments follow. All fragments of a datagram carry the same identification. IPv4 header is 20 bytes minimum, 60 maximum. Total length includes the header.

Routing, IPv4, CIDR, NAT: Private and NAT = Private: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. NAT rewrites the private source to a public address, remembered in a table. ARP: IP to MAC on the local link. DHCP gives the IP. ICMP carries errors such as unreachable and echo.

TCP, UDP, application ports: TCP control = Sequence number counts bytes. Ack is the next byte expected. Window is the receiver's free buffer. UDP has no connection, no retransmission, no congestion window. Socket is (IP, port) at one end. A TCP connection is a pair of sockets.

TCP, UDP, application ports: Congestion window = Slow start: cwnd doubles each RTT until ssthresh. Then congestion avoidance: $cwnd \pm 1$ MSS per RTT. Timeout: ssthresh = $cwnd / 2$, $cwnd = 1$ MSS. Reno triple duplicate ACK: fast retransmit, ssthresh =

28 Oct Budhvar**Revision: DBMS aur Networks (contd)**

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

cwnd/2, then fast recovery. Tahoe sets cwnd to 1 on any loss signal. Reno keeps the pipe fuller on triple ACK.

TCP, UDP, application ports: Ports = DNS 53, DHCP 67/68, HTTP 80, HTTPS 443, SMTP 25, FTP 20 data and 21 control. DNS is usually UDP, TCP if the response is large. SMTP sends mail. HTTP fetches the web.

TCP, UDP, application ports: Flow vs congestion = Flow control: receiver window, do not overrun the receiver. Congestion control: cwnd, do not overrun the network. Send window = min(receiver window, cwnd). Silly window happens when the receiver opens a tiny window. Nagle and delayed ACK interact.

06:00-07:20 1h 20m ERROR LOG**Forms aur clause order**

1NF se BCNF ek-ek line. SELECT ka clause order. TCP vs UDP ki chhe lines. Subnet ke steps.

07:20-07:35 15 min BREAK**Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50 1h 15m PRACTICE**8 SQL + 8 subnet**

Timed. SQL 25 minute, subnet 25 minute, review 25 minute. Inhe 25 sawal gino.

25 technical MCQ**08:50-10:00 1h 10m BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00 3h OFFICE**Office, subah**

Kaam. 13:00 par ER, algebra, SQL, normal forms, Indexing, transactions, concurrency, Delay, framing, error, MAC, Routing, IPv4, CIDR, NAT, TCP, UDP, application ports ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20 20 min OFFICE DESK**Lunch: ER, algebra, SQL, normal forms, Indexing, transactions, concurrency, Delay, framing, error, MAC, Routing, IPv4, CIDR, NAT, TCP, UDP, application ports**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. ER, algebra, SQL, normal forms, Indexing, transactions, concurrency, Delay, framing, error, MAC, Routing, IPv4, CIDR, NAT, TCP, UDP, application ports.

6 technical MCQ**13:20-16:30 3h 10m OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 12 min OFFICE DESK**Chai: formula band karke**

12 minute, zaroori. Sheet band. ER, algebra, SQL, normal forms, Indexing, transactions, concurrency, Delay, framing, error, MAC, Routing, IPv4, CIDR, NAT, TCP, UDP, application ports mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 2h 8m OFFICE**Office, shaam**

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 10 min OFFICE DESK**Nikalte hue 5 formulas**

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. ER, algebra, SQL, normal forms, Indexing, transactions, concurrency, Delay, framing, error, MAC, Routing, IPv4, CIDR, NAT, TCP, UDP, application ports.

28 Oct Budhvar**Revision: DBMS aur Networks (contd)**

Padhai 5h 57m | 66 technical 25 reasoning 15 GA

19:00-20:00**1h****BREAK****Ghar aur khana**

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25**25 min****GA****Current, 15**

Apni 10 headings mein se aaj 15 ek-line sawal banao aur khud solve karo.

15 GA**20:25-21:40****1h 15m****PRACTICE****35 MCQ, DBMS + CN**

Normalization, transaction, CRC, routing, application protocols. Hafta 2-3 ki galtiyan beech mein ghol do.

35 technical MCQ**21:40-21:50****10 min****BREAK****Das minute**

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35**45 min****REASONING****Mix 25**

Puzzle ya seating ek, baaki series aur coding.

25 reasoning**22:35-22:50****15 min****ERROR LOG****Jo form ab bhi uljha hai**

Sirf ek normal form. Example ke saath. 23:00 band.

29 Oct Guruvar**Revision: TOC, Compiler, Maths**

Padhai 5h 57m | 56 technical 25 reasoning 15 GA

Official syllabus lines

Paper-I + II 14.3.4 Revision: Second pass: Theory of Computation, Compiler Design, Engineering Mathematics

Paper-I 14.2.1 Revision: Second pass: reasoning - analogy through analytical functions

Paper-I 14.2.2 Revision: Second pass: history, culture, geography, economy, polity, science, current events

Formula sheet for this day

Theory of Computation: DFA aur NFA = DFA: $(Q, \Sigma, \delta, q_0, F)$, $\delta: Q \times \Sigma \rightarrow Q$, exactly one move. NFA: δ returns a set. Subset construction builds the DFA. Epsilon-NFA allows an empty move. Regular expressions, NFA, and DFA describe the same languages.

Theory of Computation: Pumping, regular = Exists p . For every w in L with $|w| \geq p$: $w = xyz$, $|xy| \leq p$, $|y| \geq 1$, and xy^kz is in L for every $k \geq 0$. Use it to prove a language is not regular. It cannot prove that a language is regular.

Theory of Computation: CFG aur PDA = CFG: (V, Σ, R, S) . Every context-free language has a PDA. Pumping for CFL: $w = uvxyz$, $|vxy| \leq p$, $|vy| \geq 1$, and uv^kxv^kyz stays in L . A grammar is ambiguous if some string has two leftmost derivations or two parse trees.

Theory of Computation: TM aur decidability = Decidable: a TM halts with yes or no on every input. Recognizable: it halts on yes, and may loop on no. Halting problem is recognizable and undecidable. Type 3 regular, type 2 context-free, type 1 context-sensitive, type 0 unrestricted.

Compiler Design: Phases = Lex $\rightarrow$ parse $\rightarrow$ semantic $\rightarrow$ intermediate code $\rightarrow$ optimize $\rightarrow$ target code. The symbol table is used across phases. Lexemes become tokens. Parse builds the tree. Semantic checks types.

Compiler Design: FIRST = FIRST(X) holds terminals that can begin a string derived from X . If X derives epsilon, epsilon is in FIRST(X). FIRST of a sequence: take FIRST of each symbol until one does not derive epsilon.

Compiler Design: FOLLOW = FOLLOW(A) holds terminals that can sit just after A in some sentential form. $\$$ is in FOLLOW of the start symbol. If $A \rightarrow \alpha B \beta$ and β derives epsilon, FOLLOW(A) joins FOLLOW(B).

Compiler Design: LL(1) test = For $A \rightarrow \alpha \mid \beta$: FIRST(α) and FIRST(β) are disjoint. If α derives epsilon, FIRST(β) and FOLLOW(A) are disjoint. One lookahead. A conflict means the grammar is not LL(1), even if the language has some other LL(1) grammar.

Compiler Design: Shift-reduce = Actions: shift, reduce, accept, error. Shift/reduce and reduce/reduce are conflicts. Bottom-up. The handle is the right-hand side that is reduced.

Compiler Design: Local dataflow = Constant propagation: every reaching definition assigns the same constant. Liveness: a value is used on some path before the next definition. CSE: the same expression is computed again and operands are unchanged.

Three-address code has at most one operator: $x = y \text{ op } z$.

Discrete mathematics: Implication = $p \rightarrow q$ is equivalent to $(\neg p) \vee q$. Contrapositive $\neg q \rightarrow \neg p$ is equivalent. Converse $q \rightarrow p$ is not. Inverse is $\neg p \rightarrow \neg q$. It matches the converse, not the original.

Discrete mathematics: Set sizes = $|A \cup B| = |A| + |B| - |A \cap B|$. De Morgan: $(A \cup B)' = A' \cap B'$. Three sets: add singles, subtract pairs, add back the triple intersection.

Discrete mathematics: Relations = Equivalence: reflexive, symmetric, transitive. Partial order: reflexive, antisymmetric, transitive. A partial order is a poset. A lattice gives every pair a join and a meet.

Discrete mathematics: Group = Closure, associativity, identity, inverse. Abelian if also commutative. Lagrange: subgroup order divides $|G|$. A monoid has closure, associativity, identity, and may lack inverses.

Discrete mathematics: Graph counts = Sum of degrees = $2|E|$. Tree: connected, $|E| = |V| - 1$. Complete graph edges = $n(n-1)/2$.

Bipartite iff no odd cycle. Handshaking says the number of odd-degree vertices is even.

Discrete mathematics: Counting = $nPr = n! / (n-r)!$. $nCr = n! / (r! (n-r)!)$. Sum of geometric series $a + ar + \dots + ar^{n-1} = a(r^n - 1) / (r - 1)$, $r \neq 1$. Recurrence: solve the characteristic equation for linear constant coefficients. Generating function $A(x) = \sum a_n x^n$.

Linear algebra: Determinant = $\det(AB) = \det(A) \det(B)$. $\det(A^T) = \det(A)$. $\det(cA) = c^n \det(A)$ for an n by n matrix. Inverse exists iff $\det \neq 0$. $(AB)^{-1} = B^{-1} A^{-1}$.

Linear algebra: Eigen = $A v = \lambda v$, $v \neq 0$. Characteristic equation $\det(A - \lambda I) = 0$. Trace = sum of eigenvalues. $\det =$ product of eigenvalues. A symmetric real matrix has real eigenvalues and orthogonal eigenvectors.

Linear algebra: LU = $A = L U$. Solve $L y = b$ by forward substitution, then $U x = y$ by back substitution. $P A = L U$ when a pivot swap is required. L is unit lower triangular in the usual form.

Linear algebra: Linear system = Unique solution if $\text{rank}(A) = \text{rank}([A|b]) = n$. Infinite if $\text{rank}(A) = \text{rank}([A|b]) < n$. None if $\text{rank}(A) < \text{rank}([A|b])$. Rank is the number of nonzero rows in row-echelon form. Rank-nullity: $\text{rank} + \text{nullity} = \text{number of columns}$.

Calculus: Limits to remember = $\sin x / x \rightarrow 1$ as $x \rightarrow 0$. $(1+x)^{1/x} \rightarrow e$ as $x \rightarrow 0$. $(1+1/n)^n \rightarrow e$. Continuity at a : limit equals the function value. Differentiable implies continuous. The converse is false.

Calculus: Derivatives = $(uv)' = u'v + uv'$. $(u/v)' = (u'v - uv') / v^2$. Chain: $dy/dx = dy/du * du/dx$. $x^n \rightarrow n x^{n-1}$. $e^x \rightarrow e^x$. $\ln x \rightarrow 1/x$. $\sin \rightarrow \cos$. $\cos \rightarrow -\sin$. A critical point has $f' = 0$ or f' undefined.

Calculus: MVT aur extrema = Rolle: $f(a)=f(b)$ implies some c with $f'(c)=0$. MVT: $f'(c) = (f(b)-f(a))/(b-a)$. Local max: $f'=0$ and $f''<0$. Local

29 Oct Guruvar**Revision: TOC, Compiler, Maths (contd)**

Padhai 5h 57m | 56 technical 25 reasoning 15 GA

min: $f''=0$ and $f''>0$. If $f''=0$ the test fails. Check the sign of f' around the point.

Calculus: Integrals = $\int x^n dx = x^{(n+1)/(n+1)} + C$, $n \neq -1$. $\int dx/x = \ln|x| + C$. $\int e^x dx = e^x + C$. Parts: $\int u dv = u v - \int v du$. Definite integral is net area. Swap limits and the sign flips.

Probability and statistics: Conditional aur Bayes = $P(A|B) = P(A \text{ and } B) / P(B)$. Bayes: $P(A|B) = P(B|A) P(A) / P(B)$. Total: $P(B) = \sum P(B|A_i) P(A_i)$. Independent means $P(A \text{ and } B) = P(A)P(B)$, so $P(A|B)=P(A)$.

Probability and statistics: Mean aur spread = $E[X] = \sum x p(x)$. $\text{Var} = E[X^2] - (E[X])^2$. $\text{SD} = \sqrt{\text{Var}}$. Median splits the ordered list. Mode is the most frequent value. For grouped data the mean uses class marks. Two modes: bimodal.

Probability and statistics: Binomial = $P(K=k) = C(n,k) p^k (1-p)^{(n-k)}$. Mean = $n p$. Variance = $n p (1-p)$. n independent trials, two outcomes, constant p .

Probability and statistics: Poisson = $P(K=k) = e^{(-\lambda)} * \lambda^k / k!$. Mean = variance = λ . Use for rare events in a fixed interval. It is the limit of binomial when n is large and p is small, $\lambda = n p$.

Probability and statistics: Uniform, exponential, normal = Uniform $[a,b]$: mean $(a+b)/2$, variance $(b-a)^2 / 12$. Exponential: mean $1/\lambda$, variance $1/\lambda^2$, memoryless. Normal $z = (x - \mu) / \sigma$. Exponential pdf is $\lambda e^{(-\lambda x)}$ for $x \geq 0$.

Standard normal has $\mu 0$ and $\sigma 1$.

Reasoning patterns: Letter position = $A=1 \dots Z=26$. Opposite pairs sum to 27: A-Z, B-Y, C-X. A jump of +1, -1, +2, -2 is the first pattern to test.

Reasoning patterns: Number series = Check first difference, then second difference, then alternate terms, then $*n + k$ or $n^2 + k$. If two series are interleaved, split odd and even positions.

Reasoning patterns: Coding = Letter shift, reverse the word, square, or position sum. If it does not show in 20 seconds, mark and return. Number coding often uses the letter positions added or multiplied.

Reasoning patterns: Relation aur direction = Draw the family. One generation is one row. Directions: sketch, do not turn in the head. A right turn is clockwise. Analogy is the same relation on the second pair. Classification: three share a rule, one does not.

06:00-07:20**1h 20m****ERROR LOG****Teen sheets, yaad se**

Language classes ka ek example each. FIRST/FOLLOW ke steps. Bayes, mean/variance, relation properties, 2x2 eigen. Teesra page shuru mat karo.

07:20-07:35**15 min****BREAK****Chhoti break**

Chai. Agla set pehle se khol ke rakho, taaki agle 75 minute sirf solve aur review mein jaayein.

07:35-08:50**1h 15m****PRACTICE****25 MCQ - TOC aur Compiler**

Language identify, phase identify, ek chhota FIRST set.

25 technical MCQ**08:50-10:00****1h 10m****BREAK****Office ke liye nikalna**

9:15 tak nikal jao. Bag mein ek formula sheet. Commute par gaana, ya GA ke 10 flashcard. Naya chapter nahi.

10:00-13:00**3h****OFFICE****Office, subah**

Kaam. 13:00 par Theory of Computation, Compiler Design, Discrete mathematics, Linear algebra, Calculus, Probability and statistics, Reasoning patterns ki sheet khulegi. Abhi naya topic nahi.

13:00-13:20**20 min****OFFICE DESK****Lunch: Theory of Computation, Compiler Design, Discrete mathematics, Linear algebra, Calculus, Probability and statistics, Reasoning patterns**

20 minute, zaroori. Formula tab mein aaj ki lines kholo. Har formula ek baar bol ke tick karo. Phir usi topic ke 6 chhote MCQ, solution ek line. Theory of Computation, Compiler Design, Discrete mathematics, Linear algebra, Calculus, Probability and statistics, Reasoning patterns.

6 technical MCQ

29 Oct **Guruvar**

Revision: TOC, Compiler, Maths (contd)

Padhai 5h 57m | 56 technical 25 reasoning 15 GA

13:20-16:30 **3h 10m** **OFFICE**

Office, dopahar

Dopahar ka kaam. Sheet band. Agli yaad chai ke break par.

16:30-16:42 **12 min** **OFFICE DESK**

Chai: formula band karke

12 minute, zaroori. Sheet band. Theory of Computation, Compiler Design, Discrete mathematics, Linear algebra, Calculus, Probability and statistics, Reasoning patterns mein se 8 formulas yaad se bolo. Jo na aaye use tick mat karo.

16:42-18:50 **2h 8m** **OFFICE**

Office, shaam

Shaam ka kaam. 18:50 par pocket card, phir seedha ghar.

18:50-19:00 **10 min** **OFFICE DESK**

Nikalte hue 5 formulas

10 minute, zaroori. Jo lunch par nahi aaye unme se 5 line error copy mein. Raat ka slot yahin se shuru hoga. Theory of Computation, Compiler Design, Discrete mathematics, Linear algebra, Calculus, Probability and statistics, Reasoning patterns.

19:00-20:00 **1h** **BREAK**

Ghar aur khana

Ek ghanta buffer. Padhai 20:00 baje. Thake hue dimaag par naya topic tabhi jab slot mein likha ho, usse pehle nahi.

20:00-20:25 **25 min** **GA**

Ek page polity + itihaas

15 rapid. Naya fact nahi.

15 GA

20:25-21:40 **1h 15m** **PRACTICE**

25 MCQ - Maths

Relation, eigen, Bayes, distribution identify. Calculus ke do sawal. Jo formula na yaad aaye use sheet par last line mein likho.

25 technical MCQ

21:40-21:50 **10 min** **BREAK**

Das minute

Paani aur stretch. Phone dusre kamre mein.

21:50-22:35 **45 min** **REASONING**

Mix 25

Poora mix, 25 minute. Yeh Paper-I ki rehearsal hai.

25 reasoning

22:35-22:50 **15 min** **ERROR LOG**

Do page band

Formula sheet ko do page par rok do. Kal subah yahi sheet hai, naya likhna nahi. 23:00 band.

30 Oct Shukravar**Paper-II mock**

Padhai 4h 52m | 110 technical 0 reasoning 0 GA

Aaj naya sawal subah ko sirf 10, confidence ke liye. Asli kaam raat ki mock hai. Mock ke baad poora solution mat likho.

Official syllabus lines

Paper-I + II 14.3.4 Mock: Paper-II mock: every Part-D topic, 100 questions, -1

Formula sheet for this day

High-yield, mock week: De Morgan, flip-flop, 2's = $(A+B)'=A'B'$. JK: $Q+=JQ'+K'Q$. D: $Q+=D$. T: $Q+=T \text{ xor } Q$. 2's: invert, add 1. n-bit signed: $-2^{(n-1)}..2^{(n-1)}-1$. IEEE single bias 127.

High-yield, mock week: CPU, pipeline, cache = Time = $IC \cdot CPI \cdot \text{cycle}$. Speedup $\rightarrow$ k stages. AMAT = hit time + miss rate*penalty. Tag = bits - index - offset. EAT form depends on whether a miss also pays the cache time.

High-yield, mock week: Master, MST, shortest path = Master: $c=\log_b a$, three cases on f vs n^c . MST V-1 edges. Dijkstra only if weights ≥ 0 . Bellman-Ford V-1 passes, $O(VE)$. Negative cycle: one extra Bellman-Ford pass still updates.

High-yield, mock week: OS numbers = TAT=CT-AT. WT=TAT-BT. Need=Max-Allocation. Offset bits= $\log_2(\text{page size})$. Fault EAT= $(1-p) \cdot ma + p \cdot \text{service}$. Deadlock needs all four conditions.

High-yield, mock week: DBMS tests = 3NF: $X \rightarrow A$ means X superkey or A prime. BCNF: X superkey. Lossless if intersection is a key of one side. Conflict serializable iff precedence graph has no cycle. Strict 2PL holds write locks until commit.

High-yield, mock week: Network numbers = $Tt=L/R$. $Tp=d/v$. Stop-and-wait $U=1/(1+2a)$, $a=Tp/Tt$. Hosts= $2^{(32-p)}-2$. Shannon $C=B \log_2(1+S/N)$. Offset bytes = field*8. Send window = min(receiver window, cwnd). Timeout sets cwnd to 1.

High-yield, mock week: TOC aur compiler = Regular pump: $|xy| \leq p$, $|y| \geq 1$. LL(1): FIRST sets disjoint, and epsilon pulls in FOLLOW. Live: used before the next definition. Halting problem is undecidable.

High-yield, mock week: Math numbers = Bayes $P(A|B)=P(B|A)P(A)/P(B)$. Binomial mean np, variance $np(1-p)$. Poisson mean=variance= λ . Eigen: $\det(A-\lambda I)=0$. MVT slope= $(f(b)-f(a))/(b-a)$. Uniform variance $(b-a)^2/12$. Exponential mean $1/\lambda$.

06:00-07:00**1h****ERROR LOG****Sirf do-page formula sheet**

Naya topic nahi, nayi sheet nahi. Ek baar poori sheet. Jo line na samajh aaye use mock ke baad ke liye mark karo, abhi research nahi.

07:00-07:20**20 min****BREAK****Nashta**

Halka. Mock raat ko hai, subah thakna nahi.

07:20-08:00**40 min****PRACTICE****10 aasaan MCQ**

Wahi topics jo aate hain. Analysis lamba nahi. Yeh confidence hai, syllabus nahi.

10 technical MCQ**08:00-10:00****2h****BREAK****Taiyaari aur nikalna**

Aaj subah chhoti hai. 9:15 tak nikalna waise hi.

10:00-13:00**3h****OFFICE****Office, subah**

Kaam shuru. Aaj mock hai. Beech mein naya chapter nahi. Lunch par sirf high-yield sheet.

13:00-13:20**20 min****OFFICE DESK****Lunch: high-yield formula, 20 minute**

High-yield, mock week. Har line ek baar padho aur tick karo. Naya MCQ nahi. Jo line atke use star, research nahi.

13:20-16:30**3h 10m****OFFICE****Office, dopahar**

Dopahar ka kaam. Sheet band. Mock ke liye dimaag khali rakho.

30 Oct Shukravar

Paper-II mock (contd)

Padhai 4h 52m | 110 technical 0 reasoning 0 GA

16:30-16:42 12 min OFFICE DESK

Chai: 12 minute, band karke

Sheet band. Star wali lines ek baar bol ke dekho. Na aaye to chhod do.

16:42-18:50 2h 8m OFFICE

Office, shaam

Shaam ka kaam. 18:50 se pehle notes mat kholna.

18:50-19:00 10 min OFFICE DESK

Nikalte hue 5 line

Pocket card par sirf 5 formulas. Phir pen band. Raat ki mock isi se garam hogi, naya topic nahi.

19:00-20:00 1h BREAK

Khana, halka

Bhaari khana nahi. 20:00 baje timer.

20:00-22:00 2h MOCK

Paper-II: 100 sawal, 120 minute

Sirf CS & IT. Har galat par -1, yaani teen galat ek sahi ko kha jaate hain. Jo 40 second mein na aaye use chhodo. OMR jaisa: ek baar chuna, bas. Section ka order syllabus jaisa mat socho - pehle wahi chapter jo revision mein mazboot tha, taaki attempt jud sake.

100 technical MCQ

22:00-22:30 30 min ERROR LOG

Sirf topic count

Kitne attempt, kitne galat, kaun se paanch topics. Solution kal dopehar ko, Paper-I ke analysis ke saath. 22:30 ke baad band.

31 Oct Shanivar

Paper-I mock aur band

Padhai 7h 50m | 100 technical 75 reasoning 70 GA

Official syllabus lines

Paper-I 14.2 Mock: Paper-I mock: reasoning, general awareness, and Part-D, 200 questions, -0.25

Formula sheet for this day

High-yield, mock week: De Morgan, flip-flop, $2's = (A+B)' = A'B'$. JK: $Q+ = JQ' + K'Q$. D: $Q+ = D$. T: $Q+ = T \text{ xor } Q$. 2's: invert, add 1. n-bit signed: $-2^{(n-1)} \dots 2^{(n-1)} - 1$. IEEE single bias 127.

High-yield, mock week: CPU, pipeline, cache = Time = $IC * CPI * cycle$. Speedup $\rightarrow k$ stages. AMAT = hit time + miss rate * penalty. Tag = bits - index - offset. EAT form depends on whether a miss also pays the cache time.

High-yield, mock week: Master, MST, shortest path = Master: $c = \log_b a$, three cases on f vs n^c . MST V-1 edges. Dijkstra only if weights ≥ 0 . Bellman-Ford V-1 passes, $O(VE)$. Negative cycle: one extra Bellman-Ford pass still updates.

High-yield, mock week: OS numbers = $TAT = CT - AT$. $WT = TAT - BT$. Need = Max-Allocation. Offset bits = $\log_2(\text{page size})$. Fault $EAT = (1-p) * ma + p * service$. Deadlock needs all four conditions.

High-yield, mock week: DBMS tests = 3NF: $X \rightarrow A$ means X superkey or A prime. BCNF: X superkey. Lossless if intersection is a key of one side. Conflict serializable iff precedence graph has no cycle. Strict 2PL holds write locks until commit.

High-yield, mock week: Network numbers = $Tt = L/R$. $Tp = d/v$. Stop-and-wait $U = 1/(1+2a)$, $a = Tp/Tt$. Hosts = $2^{(32-p)-2}$. Shannon $C = B \log_2(1+S/N)$. Offset bytes = field * 8. Send window = min(receiver window, cwnd). Timeout sets cwnd to 1.

High-yield, mock week: TOC aur compiler = Regular pump: $|xy| \leq p$, $|y| \geq 1$. LL(1): FIRST sets disjoint, and epsilon pulls in FOLLOW. Live: used before the next definition. Halting problem is undecidable.

High-yield, mock week: Math numbers = Bayes $P(A|B) = P(B|A)P(A)/P(B)$. Binomial mean np, variance $np(1-p)$. Poisson mean = variance = lambda. Eigen: $\det(A - \lambda I) = 0$. MVT slope = $(f(b) - f(a))/(b - a)$. Uniform variance $(b-a)^2/12$. Exponential mean $1/\lambda$.

Reasoning patterns: Letter position = A=1 ... Z=26. Opposite pairs sum to 27: A-Z, B-Y, C-X. A jump of +1, -1, +2, -2 is the first pattern to test.

Reasoning patterns: Number series = Check first difference, then second difference, then alternate terms, then $*n + k$ or $n^2 + k$. If two series are interleaved, split odd and even positions.

Reasoning patterns: Coding = Letter shift, reverse the word, square, or position sum. If it does not show in 20 seconds, mark and return. Number coding often uses the letter positions added or multiplied.

Reasoning patterns: Relation aur direction = Draw the family. One generation is one row. Directions: sketch, do not turn in the head. A right turn is clockwise. Analogy is the same relation on the second pair. Classification: three share a rule, one does not.

08:00-10:00

2h

MOCK

Paper-I: 200 sawal, 120 minute

Negative -0.25. Order: pehle 100 technical, phir 50 reasoning, phir 50 GA. Technical fresh dimaag se. 20 second mein na aaye to mark karke aage. GA mein andaza tabhi jab do options tak simat jaaye.

100 technical MCQ | 50 reasoning | 50 GA

10:00-10:20

20 min

BREAK

Break

Score dekho, abhi solution nahi.

10:20-12:00

1h 40m

ERROR LOG

Dono mock ka analysis

Kal raat ka Paper-II aur aaj ka Paper-I. Topic-wise sahi/galat. 50% se neeche wale topics ki list, zyada se zyada chaar.

12:00-14:30

2h 30m

BREAK

Khana aur aaram

Yeh aakhri aaram hai. Naya chapter is gap mein bhi nahi.

14:30-16:30

2h

ERROR LOG

Sirf kamzor topics ke galat sawal

Accuracy 50% se neeche wale topics. Sahi concept ek line. Naya prashna-bank nahi kholna.

16:30-17:15

45 min

REASONING

Jo mock mein galat hue

25 sawal unhi types ke. Naya type shuru nahi karna.

25 reasoning

31 Oct Shanivar

Paper-I mock aur band (contd)

Padhai 7h 50m | 100 technical 75 reasoning 70 GA

17:15-18:00 45 min GA

Rapid 20

Polity aur current headings. Jo fact mock mein galat hua wahi pehle.

20 GA

18:00-18:40 40 min ERROR LOG

Do page aur ek blacklist

Formula sheet ek baar. Alag se das cheezein jinhe dobara galat nahi karna. Iske baad yeh crash cycle band hai.

Formula sheet

Office lunch uses the day's sheet. Tick a line only when you can say it closed.

Digital Logic

De Morgan: $(A+B)' = A'B'$ $(AB)' = A'+B'$. NAND is OR of complements. NOR is AND of complements.

XOR aur XNOR: $A \text{ xor } B = A'B + AB'$ $XNOR = AB + A'B'$. XOR is 1 when bits differ. XNOR is 1 when bits match.

Absorption: $A + AB = A$ $A(A+B) = A$ $A + A'B = A + B$. Drop the larger product term when one term contains the other.

2's complement: Invert every bit, then add 1. Range of n bits: $-2^{(n-1)} \dots 2^{(n-1)}-1$. Unsigned range is $0 \dots 2^n - 1$. One's complement is invert only.

r's complement: $(r-1)$'s complement = $(r^n - 1) - N$. r's complement = $r^n - N$. For decimal, 9's then add 1 gives 10's complement.

Flip-flop next state: SR (SR=0): $Q+ = S + R'Q$. JK: $Q+ = JQ' + K'Q$. D: $Q+ = D$. T: $Q+ = T \text{ xor } Q$. SR = 11 is invalid. JK = 11 toggles. T = 1 toggles.

Counter aur MUX: n flip-flops, ripple up-counter: $\text{mod } 2^n$, $f_{\text{out}} = f_{\text{in}} / 2^n$. MUX: 2^n data, n select. Decoder: n to 2^n . A mod-10 counter needs extra reset logic. It is not a pure 4-bit free counter.

K-map group: Group size must be 1, 2, 4, 8, ... Circle power-of-two 1s (or 0s for POS). Don't-care may join a group, never a group of its own. Adjacent cells differ by one bit, including wrap of the edges.

IEEE-754 single: 1 sign, 8 exponent, 23 fraction. Bias 127. Normal value = $(-1)^s * (1.f) * 2^{(E-127)}$. E = 0 and f = 0 is zero. E = 255 and f = 0 is infinity. E = 255 and f != 0 is NaN. Double bias is 1023.

Computer Organization

CPU time: $\text{CPU time} = \text{IC} * \text{CPI} * \text{clock cycle}$. Clock cycle = $1 / \text{frequency}$. IC is instruction count. A lower CPI or a faster clock both cut time.

Pipeline speedup: No stalls, n instructions, k stages: $S = (n*k) / (k + n - 1)$. Large n: S approaches k. One stall cycle drops the ideal throughput. Efficiency = S / k .

Stall CPI: Ideal pipeline CPI = 1. Effective CPI = $1 + \text{stall cycles per instruction}$. Data hazard: forwarding removes many, not all, stalls. Control hazard: branch delay or flush.

Cache AMAT: $\text{AMAT} = \text{Hit time} + \text{Miss rate} * \text{Miss penalty}$. Miss rate = $1 - \text{hit rate}$. Two-level: $\text{AMAT} = T_{L1} + M1*(T_{L2} + M2*T_{\text{mem}})$.

Hit-rate EAT: $\text{EAT} = h*T_{\text{cache}} + (1-h)*T_{\text{memory}}$. If every access also pays the cache lookup: $\text{EAT} = h*T_c + (1-h)*(T_c+T_m)$. Use the form the question states.

Cache address split: Offset = $\log_2(\text{block bytes})$. Direct index = $\log_2(\text{blocks})$. Set index = $\log_2(\text{sets})$. Tag = address bits - index - offset. Sets = blocks / ways. Cache bytes = sets * ways * block bytes.

Effective address: Direct: EA = address field. Indirect: EA = $M[\text{address}]$. Indexed: EA = index + address. Relative: EA = PC + offset. Register-indirect: EA = register. Immediate has no memory EA. The operand sits in the instruction.

Interrupt vs DMA: Interrupt: CPU saves context and runs the handler. DMA: device moves a block; CPU is told at the end. Cycle stealing takes the bus for single transfers. Burst mode holds the bus for the block.

C, arrays, stacks, queues, lists

Array address: 1D: $\text{base} + i * \text{size}$. Row-major $A[i][j] = \text{base} + (i*\text{cols} + j)*\text{size}$. Column-major: $\text{base} + (j*\text{rows} + i)*\text{size}$. C uses row-major and 0-based indexes.

Pointer step: $p + k$ moves k elements, not k bytes. Byte move = $k * \text{sizeof}(*p)$. $a[i]$ is $*(a+i)$. $\&a[i]$ is $a+i$.

Stack aur queue: Stack: push and pop at the same end, $O(1)$. Circular queue: front and rear mod capacity, $O(1)$. Infix to postfix uses a stack. BFS uses a queue. DFS uses a stack.

Linked list: Insert or delete at a known node: $O(1)$. Search: $O(n)$. Extra memory: one pointer per node. Doubly linked delete of a known node does not need the previous walk.

Recursion cost: Time follows the recurrence of the calls. Depth of the call stack is the deepest chain, often $O(n)$ or $O(\log n)$. A missing base case overflows the stack. Tail recursion can reuse the frame.

Trees, heaps, graphs

Binary tree counts: Max nodes at level L (root level 0) = 2^L . Perfect tree height h: nodes = $2^{(h+1)} - 1$. Full tree: leaves = internal nodes + 1. Height of a single node is 0 in this sheet. Some books use 1. Match the question.

BST: Left subtree < node < right subtree. Search, insert, delete: $O(h)$. Balanced $h = \text{floor}(\log_2 n)$. Skewed $h = n-1$. Inorder of a BST is sorted order.

Binary heap index: 0-based: parent $\text{floor}((i-1)/2)$, left $2i+1$, right $2i+2$. Height $\text{floor}(\log_2 n)$. Build $O(n)$. Extract $O(\log n)$. Min-heap:

parent $\leq$ children. Heap is a complete binary tree.

Graph storage: Adjacency matrix: $O(V^2)$ space, edge check $O(1)$. Adjacency list: $O(V+E)$ space, edge check $O(\text{degree})$. Undirected edge appears twice in the list, once in the matrix triangle.

Searching, sorting, hashing

Binary search: Comparisons about $\text{floor}(\log_2 n) + 1$. Needs a sorted array. Time $\Theta(\log n)$. Lower bound of comparison sort is $\Omega(n \log n)$.

Sorting cost: Insertion, selection, bubble: worst $\Theta(n^2)$. Merge: $\Theta(n \log n)$, extra $O(n)$. Heap: $\Theta(n \log n)$, in place. Quick: average $\Theta(n \log n)$, worst $\Theta(n^2)$. Insertion best case $\Theta(n)$ on sorted input. Merge and heap are stable? Merge is stable. Heap and quick are not.

Stability: Stable: equal keys keep their input order. Merge and insertion are stable. Selection, heap, and quick are not. Counting sort is stable if the prefix scan fills from the end.

Counting and radix: Counting: $O(n + k)$ time, k is the key range. Radix: $O(d(n + k))$ for d digits. Use when keys are integers in a known range, not for arbitrary comparisons.

Hash load: $\alpha = n / m$. Chaining, unsuccessful search about $1 + \alpha$. Successful about $1 + \alpha/2$. Open addressing degrades as α approaches 1. Keep α well below 1.

Greedy, DP, divide-and-conquer

Master theorem: $T(n) = a T(n/b) + f(n)$. Let $c = \log_b(a)$. If $f = O(n^{c-\epsilon})$: $\Theta(n^c)$. If $f = \Theta(n^c)$: $\Theta(n^c \log n)$. If $f = \Omega(n^{c+\epsilon})$ and $a f(n/b) \leq k f(n)$ for $k < 1$: $\Theta(f(n))$. ϵ is a positive constant. Merge sort: $a=2$, $b=2$, $f=n$, $c=1$, case 2.

Divide and conquer: Split, solve, combine. Binary search, merge sort, quicksort, Strassen. Cost is the recurrence, then Master theorem. Not every split is equal. Quicksort's worst split is 1 and $n-1$.

Greedy choice: Feasible choice that looks best now, plus optimal substructure. Fractional knapsack: sort by value/weight. 0/1 knapsack is not greedy. It is DP. Activity selection: pick the finish-earliest.

DP shape: Optimal substructure and overlapping subproblems. 0/1 knapsack: $dp[i][w] = \max(dp[i-1][w], v_i + dp[i-1][w-w_i])$. Fibonacci naive recursion is exponential. With a table it is $O(n)$.

Graph algorithms

BFS and DFS: Time $O(V+E)$ on a list. BFS gives fewest edges from the source in an unweighted graph. DFS gives finishing times. Topological sort: decreasing finish time, and only on a DAG.

MST: Kruskal: sort edges, add if no cycle, $O(E \log E)$. Prim: grow one tree, $O(E \log V)$ with a heap. MST has $V-1$ edges. Cut property: lightest edge across a cut is in some MST. Negative weights are allowed. Negative cycles are not a concept here.

Dijkstra: Non-negative weights. Heap version $O((V+E) \log V)$. $\text{Dist}[v] = \min(\text{dist}[v], \text{dist}[u] + w(u,v))$ when u is settled. A negative edge can make a settled node wrong. Use Bellman-Ford then.

Bellman-Ford and Floyd: Bellman-Ford: relax all edges $V-1$ times, $O(VE)$. One more relax means a negative cycle. Floyd: $dp[k][i][j] = \min(dp[i][j], dp[i][k] + dp[k][j])$, $O(V^3)$. Floyd allows negative edges, not a negative cycle.

Processes, sync, deadlock

TAT, waiting, response: $\text{TAT} = \text{completion} - \text{arrival}$. $\text{Waiting} = \text{TAT} - \text{burst}$. $\text{Response} = \text{first CPU start} - \text{arrival}$. $\text{CPU utilization} = \text{busy time} / \text{total time}$. $\text{Throughput} = \text{jobs finished} / \text{time}$.

Scheduling pick: FCFS: simple, convoy. SJF: least average waiting if all arrive together, non-preemptive. SRTF: preemptive SJF. RR: quantum q , more switches as q shrinks. Priority can starve unless aging raises the waiting job.

Banker's need: $\text{Need}[i] = \text{Max}[i] - \text{Allocation}[i]$. Safe if a sequence exists: Work starts as Available, and each picked process has $\text{Need} \leq \text{Work}$, then $\text{Work} += \text{Allocation}$. An unsafe state is not a deadlock yet. It might become one.

Deadlock four: Mutual exclusion, hold and wait, no preemption, circular wait. All four together. Break any one condition and deadlock cannot form. Prevention vs avoidance vs detection.

Semaphore: Wait (P): while $S \leq 0$ loop; $S = S-1$. Signal (V): $S = S+1$. Binary semaphore is 0 or 1. Counting semaphore tracks a pool. Mutex lock is not a signal to a different process by itself. A semaphore can.

Memory and file systems

Page bits: $\text{Offset bits} = \log_2(\text{page size})$. $\text{Page number} = \text{logical address bits} - \text{offset bits}$. $\text{Frames} = \text{physical bytes} / \text{page size}$. Logical pages and physical frames are the same size.

TLB EAT: Single-level page table: $\text{EAT} = h * (T_{\text{tlb}} + T_{\text{mem}}) + (1-h) * (T_{\text{tlb}} + 2 * T_{\text{mem}})$. A miss reads the page table in memory, then the data. Two memory reads.

Page fault time: $\text{EAT} = (1-p) * \text{memory access} + p * \text{page-fault service}$. p is the fault probability. Service includes disk. It dominates

memory.

Replacement: FIFO can show Belady: more frames, more faults. Optimal: replace the page used farthest in the future. LRU: replace the least recently used. Optimal needs the future. It is a yardstick, not an online policy.

Fragmentation: Paging: internal fragment in the last page only. Segmentation: external fragment between variable segments. Compaction moves segments. Paging does not need it for external holes.

File allocation: Contiguous: direct, external fragment. Linked: no external fragment, slow random. Indexed: index block, random access, one extra read. Unix inode: direct blocks, then single, double, triple indirect.

ER, algebra, SQL, normal forms

Relational algebra: sigma select, pi project, rho rename, union, minus, cartesian product, join. Select keeps rows. Project keeps columns and removes duplicate rows.

Keys: Superkey uniquely identifies. Candidate key is a minimal superkey. Primary key is one chosen candidate. Foreign key references a candidate key. Prime attribute: member of some candidate key.

Normal forms: 1NF: atomic values. 2NF: no partial dependency of a non-prime on a candidate key. 3NF: for $X \rightarrow A$, X is a superkey or A is prime. BCNF: for every nontrivial $X \rightarrow A$, X is a superkey. BCNF implies 3NF. 3NF does not imply BCNF. A table can be 3NF and still have overlapping keys.

Lossless join: Decomposition of two parts is lossless if $R_1 \cap R_2$ is a key of R_1 or of R_2 . Dependency preserving: the union of the projected FDs is equivalent to F. Lossless and preserving are different tests.

SQL groups: WHERE filters rows before groups. HAVING filters groups. SELECT aggregates need GROUP BY of the other selected columns. COUNT(*) counts rows. COUNT(col) skips NULL. DISTINCT inside COUNT counts uniques.

Indexing, transactions, concurrency

B and B+ tree: Order-m node holds up to m children. Search $O(\log n)$. B+ tree stores records only in the leaves, and leaves are linked. A B+ leaf scan is the range query. Internal nodes are a sparse directory.

Index kind: Dense: one entry per record. Sparse: one entry per block, file must be ordered. Primary index is on the ordering key. Secondary index is on another field. Clustering index can be on a non-unique ordered field.

ACID: Atomic: all or nothing. Consistent: constraints hold. Isolated: concurrent effect matches some serial order. Durable: commit survives a crash. A schedule can be atomic per transaction and still not serializable.

Serializability: Conflict: two ops, different transactions, same item, at least one write. Conflict serializable iff the precedence graph has no cycle. Conflict serializable implies view serializable. The converse is false.

Two-phase lock: Growing phase, then shrinking phase. Strict 2PL holds exclusive locks until commit. 2PL guarantees conflict serializability. It can still deadlock.

Delay, framing, error, MAC

Four delays: Transmission = L/R . Propagation = d/v . Total = transmission + propagation + queue + processing. Bandwidth is R in bits per second. L is in bits. Do not mix bytes.

Stop-and-wait: $a = T_p / T_t$. Utilization = $1 / (1 + 2a)$. $2a$ is the round trip measured in transmission times. A large a wastes the link.

Sliding window: If $W < 1 + 2a$: utilization = $W / (1 + 2a)$. Else utilization = 1. W is the window in packets. Sequence space must be at least $W + 1$ for go-back-N in the simple model, and $2W$ for selective repeat.

Nyquist and Shannon: Noiseless Nyquist: $C = 2 * B * \log_2(M)$. Noisy Shannon: $C = B * \log_2(1 + S/N)$. If SNR is in dB, ratio = $10^{(dB/10)}$. M is the number of levels.

Hamming: Distance d detects up to $d - 1$ errors and corrects $\lfloor (d - 1) / 2 \rfloor$ errors. Parity adds distance. A single parity bit detects 1 and corrects 0.

CRC: Treat the frame as a polynomial. Append remainder of division by the generator. A zero remainder at the receiver means no detected error. CRC is strong detection, not correction.

Routing, IPv4, CIDR, NAT

Routing idea: Distance vector: Bellman-Ford, advertise the whole vector, count-to-infinity. Link state: flood links, then Dijkstra at each router. Split horizon and poison reverse cut some, not all, loops.

IPv4 subnet: Prefix /p uses p network bits. Host bits = $32 - p$. Usable hosts = $2^{(32 - p)} - 2$. The -2 drops network and broadcast. /31 and /32 are special and do not follow this usable-host rule.

Mask from prefix: /p mask: p ones, then zeros. Block size of the interesting octet = $256 / 2^{(\text{ones in that octet})}$. A /26 interesting octet is the 4th: 2 bits host in that octet, block 64.

Fragment offset: Offset field is in 8-byte units. Byte offset = offset * 8. MF = 1 means more fragments follow. All fragments of a datagram carry the same identification. IPv4 header is 20 bytes minimum, 60 maximum. Total length includes the header.

Private and NAT: Private: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. NAT rewrites the private source to a public address, remembered in a table. ARP: IP to MAC on the local link. DHCP gives the IP. ICMP carries errors such as unreachable and echo.

TCP, UDP, application ports

TCP control: Sequence number counts bytes. Ack is the next byte expected. Window is the receiver's free buffer. UDP has no connection, no retransmission, no congestion window. Socket is (IP, port) at one end. A TCP connection is a pair of sockets.

Congestion window: Slow start: cwnd doubles each RTT until ssthresh. Then congestion avoidance: cwnd += 1 MSS per RTT. Timeout: ssthresh = cwnd/2, cwnd = 1 MSS. Reno triple duplicate ACK: fast retransmit, ssthresh = cwnd/2, then fast recovery. Tahoe sets cwnd to 1 on any loss signal. Reno keeps the pipe fuller on triple ACK.

Ports: DNS 53, DHCP 67/68, HTTP 80, HTTPS 443, SMTP 25, FTP 20 data and 21 control. DNS is usually UDP, TCP if the response is large. SMTP sends mail. HTTP fetches the web.

Flow vs congestion: Flow control: receiver window, do not overrun the receiver. Congestion control: cwnd, do not overrun the network. Send window = min(receiver window, cwnd). Silly window happens when the receiver opens a tiny window. Nagle and delayed ACK interact.

Theory of Computation

DFA aur NFA: DFA: $(Q, \Sigma, \delta, q_0, F)$, $\delta: Q \times \Sigma \rightarrow Q$, exactly one move. NFA: δ returns a set. Subset construction builds the DFA. Epsilon-NFA allows an empty move. Regular expressions, NFA, and DFA describe the same languages.

Pumping, regular: Exists p . For every w in L with $|w| \geq p$: $w = xyz$, $|xy| \leq p$, $|y| \geq 1$, and xy^kz is in L for every $k \geq 0$. Use it to prove a language is not regular. It cannot prove that a language is regular.

CFG aur PDA: CFG: (V, Σ, R, S) . Every context-free language has a PDA. Pumping for CFL: $w = uvxyz$, $|vxy| \leq p$, $|vy| \geq 1$, and uv^kxv^kyz stays in L . A grammar is ambiguous if some string has two leftmost derivations or two parse trees.

TM aur decidability: Decidable: a TM halts with yes or no on every input. Recognizable: it halts on yes, and may loop on no. Halting problem is recognizable and undecidable. Type 3 regular, type 2 context-free, type 1 context-sensitive, type 0 unrestricted.

Compiler Design

Phases: Lex $\rightarrow$ parse $\rightarrow$ semantic $\rightarrow$ intermediate code $\rightarrow$ optimize $\rightarrow$ target code. The symbol table is used across phases. Lexemes become tokens. Parse builds the tree. Semantic checks types.

FIRST: FIRST(X) holds terminals that can begin a string derived from X . If X derives epsilon, epsilon is in FIRST(X). FIRST of a sequence: take FIRST of each symbol until one does not derive epsilon.

FOLLOW: FOLLOW(A) holds terminals that can sit just after A in some sentential form. $\$$ is in FOLLOW of the start symbol. If $A \rightarrow \alpha B \beta$ and β derives epsilon, FOLLOW(A) joins FOLLOW(B).

LL(1) test: For $A \rightarrow \alpha \mid \beta$: FIRST(α) and FIRST(β) are disjoint. If α derives epsilon, FIRST(β) and FOLLOW(A) are disjoint. One lookahead. A conflict means the grammar is not LL(1), even if the language has some other LL(1) grammar.

Shift-reduce: Actions: shift, reduce, accept, error. Shift/reduce and reduce/reduce are conflicts. Bottom-up. The handle is the right-hand side that is reduced.

Local dataflow: Constant propagation: every reaching definition assigns the same constant. Liveness: a value is used on some path before the next definition. CSE: the same expression is computed again and operands are unchanged. Three-address code has at most one operator: $x = y \text{ op } z$.

Discrete mathematics

Implication: $p \rightarrow q$ is equivalent to $(\text{not } p) \text{ or } q$. Contrapositive $\text{not } q \rightarrow \text{not } p$ is equivalent. Converse $q \rightarrow p$ is not. Inverse is $\text{not } p \rightarrow \text{not } q$. It matches the converse, not the original.

Set sizes: $|A \cup B| = |A| + |B| - |A \cap B|$. De Morgan: $(A \cup B)' = A' \cap B'$. Three sets: add singles, subtract pairs, add back the triple intersection.

Relations: Equivalence: reflexive, symmetric, transitive. Partial order: reflexive, antisymmetric, transitive. A partial order is a poset. A lattice gives every pair a join and a meet.

Group: Closure, associativity, identity, inverse. Abelian if also commutative. Lagrange: subgroup order divides $|G|$. A monoid has closure, associativity, identity, and may lack inverses.

Graph counts: Sum of degrees = $2|E|$. Tree: connected, $|E| = |V| - 1$. Complete graph edges = $n(n-1)/2$. Bipartite iff no odd cycle. Handshaking says the number of odd-degree vertices is even.

Counting: $nPr = n! / (n-r)!$. $nCr = n! / (r! (n-r)!)$. Sum of geometric series $a + ar + \dots + ar^{n-1} = a(r^n - 1) / (r - 1)$, $r \neq 1$. Recurrence: solve the characteristic equation for linear constant coefficients. Generating function $A(x) = \sum a_n x^n$.

Linear algebra

Determinant: $\det(AB) = \det(A) \det(B)$. $\det(A^T) = \det(A)$. $\det(cA) = c^n \det(A)$ for an n by n matrix. Inverse exists iff $\det \neq 0$. $(AB)^{-1} = B^{-1} A^{-1}$.

Eigen: $A v = \lambda v$, $v \neq 0$. Characteristic equation $\det(A - \lambda I) = 0$. Trace = sum of eigenvalues. $\det$ = product of eigenvalues. A symmetric real matrix has real eigenvalues and orthogonal eigenvectors.

LU: $A = L U$. Solve $L y = b$ by forward substitution, then $U x = y$ by back substitution. $P A = L U$ when a pivot swap is required. L is unit lower triangular in the usual form.

Linear system: Unique solution if $\text{rank}(A) = \text{rank}([A|b]) = n$. Infinite if $\text{rank}(A) = \text{rank}([A|b]) < n$. None if $\text{rank}(A) < \text{rank}([A|b])$. Rank is the number of nonzero rows in row-echelon form. Rank-nullity: $\text{rank} + \text{nullity} = \text{number of columns}$.

Calculus

Limits to remember: $\sin x / x \rightarrow 1$ as $x \rightarrow 0$. $(1 + x)^{1/x} \rightarrow e$ as $x \rightarrow 0$. $(1 + 1/n)^n \rightarrow e$. Continuity at a : limit equals the function value. Differentiable implies continuous. The converse is false.

Derivatives: $(uv)' = u'v + uv'$. $(u/v)' = (u'v - uv') / v^2$. Chain: $dy/dx = dy/du * du/dx$. $x^n \rightarrow n x^{n-1}$. $e^x \rightarrow e^x$. $\ln x \rightarrow 1/x$. $\sin \rightarrow \cos$. $\cos \rightarrow -\sin$. A critical point has $f' = 0$ or f' undefined.

MVT aur extrema: Rolle: $f(a)=f(b)$ implies some c with $f'(c)=0$. MVT: $f'(c) = (f(b)-f(a))/(b-a)$. Local max: $f'=0$ and $f''<0$. Local min: $f'=0$ and $f''>0$. If $f''=0$ the test fails. Check the sign of f' around the point.

Integrals: $\int x^n dx = x^{n+1}/(n+1) + C$, $n \neq -1$. $\int dx/x = \ln|x| + C$. $\int e^x dx = e^x + C$. Parts: $\int u dv = u v - \int v du$. Definite integral is net area. Swap limits and the sign flips.

Probability and statistics

Conditional aur Bayes: $P(A|B) = P(A \text{ and } B) / P(B)$. Bayes: $P(A|B) = P(B|A) P(A) / P(B)$. Total: $P(B) = \sum P(B|A_i) P(A_i)$. Independent means $P(A \text{ and } B) = P(A)P(B)$, so $P(A|B)=P(A)$.

Mean aur spread: $E[X] = \sum x p(x)$. $\text{Var} = E[X^2] - (E[X])^2$. $\text{SD} = \sqrt{\text{Var}}$. Median splits the ordered list. Mode is the most frequent value. For grouped data the mean uses class marks. Two modes: bimodal.

Binomial: $P(K=k) = C(n,k) p^k (1-p)^{n-k}$. Mean = $n p$. Variance = $n p (1-p)$. n independent trials, two outcomes, constant p .

Poisson: $P(K=k) = e^{-\lambda} * \lambda^k / k!$. Mean = variance = λ . Use for rare events in a fixed interval. It is the limit of binomial when n is large and p is small, $\lambda = n p$.

Uniform, exponential, normal: Uniform $[a,b]$: mean $(a+b)/2$, variance $(b-a)^2 / 12$. Exponential: mean $1/\lambda$, variance $1/\lambda^2$, memoryless. Normal $z = (x - \mu) / \sigma$. Exponential pdf is $\lambda e^{-\lambda x}$ for $x \geq 0$. Standard normal has $\mu = 0$ and $\sigma = 1$.

Reasoning patterns

Letter position: $A=1 \dots Z=26$. Opposite pairs sum to 27: $A-Z$, $B-Y$, $C-X$. A jump of $+1$, -1 , $+2$, -2 is the first pattern to test.

Number series: Check first difference, then second difference, then alternate terms, then $*n + k$ or $n^2 + k$. If two series are interleaved, split odd and even positions.

Coding: Letter shift, reverse the word, square, or position sum. If it does not show in 20 seconds, mark and return. Number coding often uses the letter positions added or multiplied.

Relation aur direction: Draw the family. One generation is one row. Directions: sketch, do not turn in the head. A right turn is clockwise. Analogy is the same relation on the second pair. Classification: three share a rule, one does not.

High-yield, mock week

De Morgan, flip-flop, 2's: $(A+B)' = A'B'$. JK: $Q+ = JQ' + K'Q$. D: $Q+ = D$. T: $Q+ = T \text{ xor } Q$. 2's: invert, add 1. n -bit signed: $-2^{n-1} \dots 2^{n-1}-1$. IEEE single bias 127.

CPU, pipeline, cache: $\text{Time} = \text{IC} * \text{CPI} * \text{cycle}$. Speedup $\rightarrow k$ stages. $\text{AMAT} = \text{hit time} + \text{miss rate} * \text{penalty}$. Tag = bits - index - offset. EAT form depends on whether a miss also pays the cache time.

Master, MST, shortest path: Master: $c = \log_b a$, three cases on f vs n^c . MST $V-1$ edges. Dijkstra only if weights ≥ 0 . Bellman-Ford $V-1$ passes, $O(VE)$. Negative cycle: one extra Bellman-Ford pass still updates.

OS numbers: $\text{TAT} = \text{CT} + \text{AT}$. $\text{WT} = \text{TAT} - \text{BT}$. $\text{Need} = \text{Max-Allocation}$. Offset bits = $\log_2(\text{page size})$. Fault $\text{EAT} = (1-p) * \text{ma} + p * \text{service}$. Deadlock needs all four conditions.

DBMS tests: 3NF: $X \rightarrow A$ means X superkey or A prime. BCNF: X superkey. Lossless if intersection is a key of one side. Conflict serializable iff precedence graph has no cycle. Strict 2PL holds write locks until commit.

Network numbers: $T_t = L/R$. $T_p = d/v$. Stop-and-wait $U = 1/(1+2a)$, $a = T_p/T_t$. Hosts = $2^{32-p}-2$. Shannon $C = B \log_2(1+S/N)$. Offset bytes = field*8. Send window = $\min(\text{receiver window}, \text{cwnd})$. Timeout sets cwnd to 1.

TOC aur compiler: Regular pump: $|xy| \leq p$, $|y| \geq 1$. LL(1): FIRST sets disjoint, and epsilon pulls in FOLLOW. Live: used before the

next definition. Halting problem is undecidable.

Math numbers: Bayes $P(A|B)=P(B|A)P(A)/P(B)$. Binomial mean np , variance $np(1-p)$. Poisson mean=variance= λ . Eigen: $\det(A-\lambda I)=0$. MVT slope= $(f(b)-f(a))/(b-a)$. Uniform variance $(b-a)^2/12$. Exponential mean $1/\lambda$.